

Automatic Generation of Text-Based Summaries of Intracranial EEG Seizure Events Using Large Language Models (LLMs)

by

Azhar Saparova

Submitted to the Department of Robotics and Mechatronics
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Abstract

Precise localization of epileptogenic tissue from intracranial electroencephalography (iEEG) is essential for successful epilepsy surgery, but methods used in the clinical environment remain labor-intensive, time-consuming, and difficult to standardize. This thesis presents a system for the automatic generation of reports with iEEG findings by combining signal processing, deep learning, and large language models (LLMs) techniques. The system consists of two branches: high-frequency oscillation (HFO) detection for epileptogenic zone localization and seizure detection for ictal event detection. HFO analysis was first developed with the OpenNeuro ds003498 dataset and then adapted to the HUP dataset, while seizure detection was initially explored on the SWEC-ETHZ dataset and subsequently extended to HUP. For HUP seizure detection, a CNN-transformer architecture was used and improved with model simplification, regularization, and windowing changes. Across 43 subjects, the final model achieved mean subject-level accuracy and balanced accuracy of 0.83, with median values of 0.87. The median F1-score was 0.84, while median AUROC and AUPRC reached 0.97 and 0.95, respectively, although performance remained heterogeneous across subjects. The structured outputs of both branches were exported as JSON files and used as inputs to four LLM system versions. Version 1 focused on report generation, Version 2 introduced channel-level seizure evidence and deterministic cross-modal fusion, Version 3 added schema-validated reasoning and deterministic report rendering, and Version 4 extended the system with bounded tool orchestration and planning. The framework demonstrates a step toward more automated and standardized iEEG reporting.

Thesis Supervisor: Berdakh Abibullaev
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Chapter 1

Introduction

1.1 Background

Epilepsy is a condition of brain networks characterized by a tendency for repeated seizures and the aftereffects of those seizures [2]. Initial treatment for epilepsy is pharmacological, but almost 30% of epilepsy patients are pharmacoresistant, meaning that drug therapy does not give them seizure freedom. So, it leads to epilepsy surgery as an optional treatment for those patients [3]. The purpose of epilepsy surgery is precise localization of the epileptogenic zone (EZ), resection of which will lead to seizure freedom [4]. The success of surgery significantly depends on the correct localization of EZ. In such a way, 60-90% of patients achieve seizure freedom upon complete localization [5].

Comparing existing non-invasive technologies, such as MRI, EEG, and functional neuroimaging, the intracranial electroencephalography (iEEG) is usually used for location and resectability of the SOZ [6]. The reason is that iEEG has higher spatial and temporal resolution compared to the non-invasive methods [7]. The subdural grids and depth electrodes are implanted directly to the brain areas involved in the onset and propagation of seizures to identify and characterize [8].

There are different methodologies to determine epileptogenic areas from iEEG recordings. One of the new biomarkers used for delineating epileptogenic brain areas is high-frequency oscillation (HFO) that is found in frequency range between 80–500 Hz.

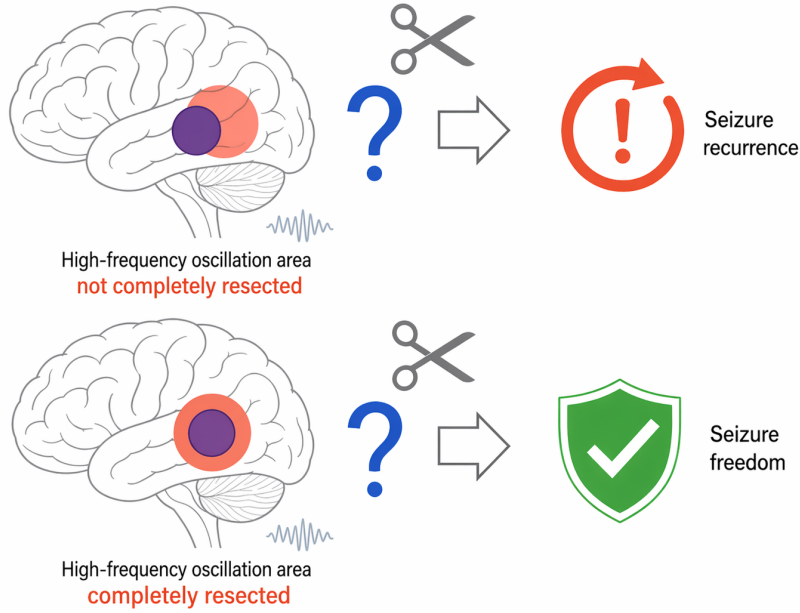


Figure 1-1: Resection of HFO area leading to seizure freedom

HFOs are also sub-classified into ripples (80–250 Hz) and fast ripples (FR, 250–500 Hz) [9]. However, there is a problem of distinguishing pathological, i.e., true HFO and physiological HFO. Physiological HFO related to sensory-motor function, cognitive activity, or the effect of high-pass filtering was found to be confused with pathological HFO during clinical testing [9]. In case of successful resection of the HFO area, patients gain seizure freedom. The visual representation explaining seizure freedom is shown in Figure 1-1.

The issue of precise HFO detection is solved with the use of automated or semi-automated detection algorithms. There are traditional methods, such as manual detection performed by looking at the monitor screen and enlarging the time scale, which were used in the Montreal Neurology Institute (MNI) [10]. In addition, automatic methods such as the line length (LL) detector, root mean square (RMS) detector, Hilbert detector, and MNI detector have been developed by different groups of researchers. Namely, LL, RMS, and Hilbert detectors take the assumption that HFOs are rare events and determine an energy threshold in the needed time segment. Meanwhile, the MNI detector computes baseline segments earlier than a detected local energy threshold [10]. Recently, there has also been a tendency in the usage of

Machine Learning (ML) and Deep Learning (DL) for the analysis of iEEG data for the localization of epileptogenic zones. DL methods are used to extract spatiotemporal patterns that associate detected events with the boundaries of epileptogenic zones and to distinguish pathological HFOs from physiological HFOs.

The same tendency can be observed for the seizure detection task. The traditional methods used handcrafted time-frequency features, such as wavelet coefficients, spectral power in canonical frequency bands, with shallow classifiers [11]. In such a way, the seizure detection task was performed with the use of support vector machines (SVM), random forests (RF), k-nearest neighbors, logistic regression, or shallow multilayer perceptron models [12, 13, 14]. However, such an approach has problems with robustness and generalization across patients; meanwhile, recent advancements in ML have led to the development of models that excel in the detection and classification of seizure events, analysis of which helps in preoperative diagnostics.

One of the most popular models is CNN, which can learn complex representations directly from raw iEEG data. The model can be used with the conversion of iEEG data into time-frequency scalograms. For example, Ma et. al. (2019) used a five-layer CNN for HFO detection by converting 80-500 Hz signal into scalograms, achieving 94.19% precision and 91.71% F1-score [15]. Another type of model is RNN, which is useful for capturing temporal dependencies as well as the LSTM architecture that deals with the vanishing gradient problem. Combination of CNN and RNN/LSTM-based models give good results because it allows them to extract local features as well as capture long-term dependencies. For example, Liu et.al. (2021) implemented a hybrid bi-branch neural network for HFO detection, combining a 1D ResNet-LSTM on bandpass-filtered signals with a 2D ResNet-CBAM on wavelet transforms, achieving high results in sensitivity, specificity, and accuracy in both intra and cross-subject tests [16]. Meanwhile, transformers have an advantage over the RNNs because of their multi-head self-attention mechanism that captures complex relationships across the entire sequence of data. Such properties are particularly useful for seizure event detection, where ictal events and interictal patterns may alter in duration and can occur at different time periods [11]. For example, Yu et al. (2023) demonstrated a model

that combines a 1D CNN and a transformer encoder with convolutional multi-head attention, outperforming prior models [17]. In general, there is a tendency in the use of hybrid models as they optimize performance and enhance clinical interpretability [11]. Transformer-based models tend to perform better compared to the CNN and RNN-based models in long-sequence tasks [11], but any task should be evaluated precisely since transformer-based models could perform poorly in small, medium-sized datasets.

Nevertheless, the detection of seizure events has several problems that mainly occur because of the variety of data from subject to subject, and even from run to run for one subject. Also, one of the problems is a huge class imbalance since seizures are rare events that appear in a limited duration of iEEG recording. Therefore, it is common practice to determine the epileptogenic zone during the interictal period, i.e., during abnormal activity happening between seizures [9].

1.2 Research Question and Aims

Even though the advances in Machine Learning (ML) and Deep Learning (DL) have enabled more automated analysis of iEEG recording, it is still a common clinical practice to use manual analysis that has a variety of problems, such as being time-consuming, labor-intensive, and subjective. Thus, there is a need in a system that will use all of the advantages of technological advancement to detect abnormal activities and report technical findings in a structured format, being also comfortable for usage among medical staff without demanding deep knowledge in ML. It is now possible to create such a system with the use of LLMs, which were initially designed for token prediction, and can now create intelligent agents that perform complex reasoning and manage multi-step tasks. In such a way, the work of iEEG analysis for seizure event detection for the purpose of pre-surgery and post-surgery diagnosis can be automated and standardized by creating automatic report generation using LLMs.

So, the following work aims to develop a completely new system for report generation from iEEG data, taking into account a sophisticated review of the existing

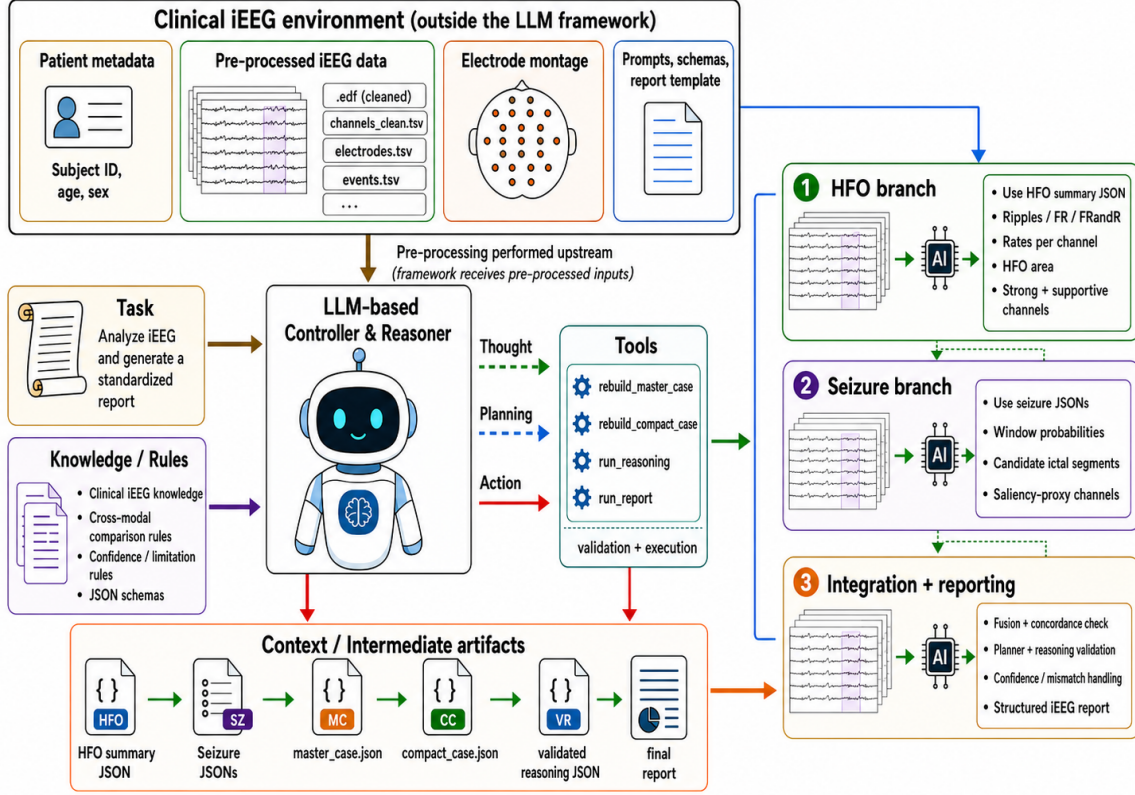


Figure 1-2: LLM system framework

works, established methods for HFO detection and seizure detection, and new advancements in DL technologies. Figure 1-2 illustrates the system framework giving a visual explanation to all development stages (HFO and seizure branches and integration with reporting), used data and the role of LLM. The system will perform a full analysis of iEEG data, starting from pre-processing till report generation, with crucial information about seizure events, applying Machine Learning techniques and LLMs. Such a system will be beneficial for hospitals reducing the manual reporting burden.

1.3 Thesis Structure

The rest of the thesis work is organized in the following way: Chapter 2 represents a literature review on methods of HFO identification and seizure detection algorithms,

and existing works in the intersection of fields among LLMs and seizure detection. The chapter concludes with a summary of research gaps that motivate the thesis work. Chapter 3 describes the used methods, which include a pre-processing pipeline, interictal HFO detection and FRandR analysis, seizure detection, and implementation of four LLM versions with the open-source Qwen3-8B model, and rubric evaluation of LLM based on three criteria. Chapter 4 reports the results of HFO detection, seizure detection, and LLM modules and makes a discussion on the main findings. The chapter includes results for seizure detection with confusion matrices, reported performance with accuracy, F1-score, precision, recall, and AUROC, as well as the 95th percentile of channels after HFO analysis. The JSON file and final report examples resulting from the LLM pipeline will also be presented in the chapter. Also, the tables for rubric evaluation based on three criteria, such as reasoning-quality rubric, faithfulness to structured evidence, and orchestration validity were represented.

Chapter 2

Literature Review

2.1 HFO detection

As it was mentioned before, there is a problem of distinguishing pathological and physiological HFO, which leads to the question of defining HFO. Clinically relevant HFO is considered to be the simultaneous occurrence of fast ripples and ripples (FRandR) since they were proven to be more specific in localizing the epileptogenic zone compared to FR or R alone [1]. One of the examples that implemented such a definition for HFO detection is the work of Dimakopoulos et.al. (2021), who assessed the reliability of the proposed detector using iEEG data recorded during non-rapid eye movement (NREM) sleep [9]. The area of HFO was computed using the HFO rate, which is calculated by dividing the HFO count per channel by duration of the epoch in minutes. Later, the spatial distribution of HFO rates across channels was analyzed for each patient, and a channel whose rates exceeded rate threshold, i.e., 95th percentile of distribution is defined as an HFO area [9]. Another work made by Fedele et.al. (2017) uses the same technique for HFO area detection, resection of which led to 13 out of 13 patients ending their therapy with seizure freedom [1]. Figure 2-1 illustrates the automated HFO analysis performed in the work of Fedele et.al., showing a presurgical MRI scan of right hippocampal sclerosis (A), MRI after implantation of 8 depth electrodes (B), and MRI resulting in long-term seizure freedom (C). Also, Figure 2-1 shows ripple co-occurring with FR (FRandR) in the green line (D), and

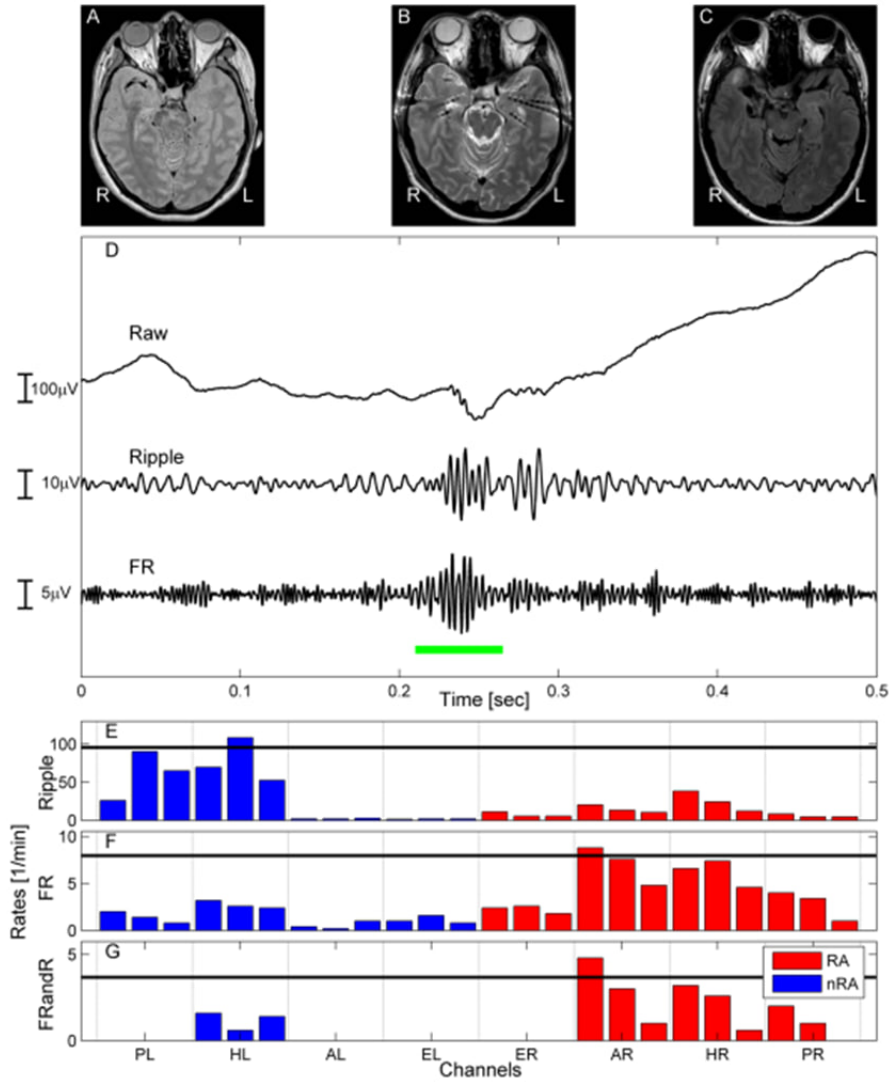


Figure 2-1: MRIs scans, iEEG recording and HFO distribution [1]

HFO distribution over channels (E, F, G) [1]. Here, channels with rates above the 95th percentile, which are illustrated as a black horizontal line, were taken to define the HFO area, where channels in red are channels inside the resected area (RA) and channels in blue are outside the RA. The ripple area was not resected (E); while, the resection of the F RandR area (G, channel AR1-AR2) led to seizure freedom [1], which proves that such detection of HFO is clinically relevant.

2.2 Seizure detection algorithms

Before reviewing seizure detection algorithms, it would be useful to briefly introduce the main deep learning models that are commonly used for iEEG analysis. These models are important because seizure detection is a time-series classification problem. The system must learn patterns in multichannel brain signals and decide whether a given time window corresponds to seizure or non-seizure activity.

2.2.1 Basic deep learning models used for iEEG analysis

A convolutional neural network (CNN) is a neural network designed to detect local patterns in data, and is composed of layers with convolutional, pooling, and final layers acting as a classifier at the end of the network [18]. Important part of the CNN network is the kernel of a specific size that helps the model to learn local features. The convolutional layers can perform convolution computations with the kernel sliding across the data matrix [19]. CNNs were originally popularized for pattern-recognition tasks such as image and document recognition, but the same idea can be applied to one-dimensional signals such as EEG and iEEG. In iEEG analysis, a CNN can learn short waveform patterns that may distinguish seizure from non-seizure activity. For a one-dimensional signal, a simple convolution operation can be written as:

$$y_t = \sum_{k=0}^{K-1} w_k x_{t+k} + b \quad (2.1)$$

where x_t is the input signal, w_k are learned filter weights, b is a bias term, K is the filter size, and y_t is the extracted feature at time t . In this way, the same learned filter is applied across different time positions of the signal, allowing the model to detect similar local patterns at different moments.

Recurrent neural networks (RNNs) are another class of models used for sequence data. Unlike CNNs, which mainly extract local patterns, RNNs process a sequence step by step and maintain a hidden state that summarizes information from previous time points. This property makes them useful for signals where temporal evolution is important. However, standard RNNs can have difficulty learning long-range temporal dependencies. Long short-term memory (LSTM) networks and gated recurrent units (GRUs) were developed to reduce this problem by using gates that control how much past information is kept or updated [20, 21].

A simplified GRU update can be described as:

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z), \quad (2.2)$$

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r), \quad (2.3)$$

$$\tilde{h}_t = \tanh(W_h x_t + U_h (r_t \odot h_{t-1}) + b_h), \quad (2.4)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t. \quad (2.5)$$

where x_t is the input at time t , h_{t-1} is the previous hidden state, h_t is the updated hidden state, z_t is the update gate, r_t is the reset gate, σ is the sigmoid function, and $\odot$ denotes element-wise multiplication. The update gate z_t controls how much new information is added to the hidden state, while the reset gate r_t controls how much previous information is forgotten. In seizure detection, this is useful because seizure patterns may evolve over time rather than appear as a single isolated waveform.

More recently, transformer-based models have been used for time-series and biomedical signal analysis. The transformer architecture was introduced by Vaswani et al. (2017) and is based on attention mechanisms rather than recurrent processing [22]. The attention mechanism allows the model to compare different parts of the input sequence and assign higher importance to the most relevant parts. The scaled dot-

product attention operation is commonly written as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (2.6)$$

where Q , K , and V are the query, key, and value matrices, and d_k is the dimensionality of the key vectors. In the context of iEEG seizure detection, attention is useful because the model can learn which time points or channels are more informative for classification. For example, when the transformer is applied across channel-level features, it can model relationships between electrodes and identify channels that contribute more strongly to the seizure prediction.

2.2.2 Comparison of existing works on seizure detection

Comparing existing works on seizure detection algorithms, it was concluded that hybrid models tend to perform better because they optimize performance and enhance clinical interpretability [11], while transformers show the highest performance. In addition, the attention mechanism can be combined with a CNN architecture that excels in establishing local features. For example, Sun et.al. (2025) implement subject-independent seizure detection combining channel-wise mixup, transformer networks, and multi-task learning [23]. In this case, channel-wise mixup is used for effective data usage to improve the performance for subject-independent training, while multi-task learning is used to improve generalization of the model by optimizing the seizure detection and the subject recognition tasks simultaneously. In the work, authors use CNNs to extract temporal features at a single-channel level, and later use a transformer network to aggregate the features from each iEEG channel [23]. The authors used SWEC-ETHZ and HUP datasets, achieving an AUC score of 0.97 and 0.95, respectively [23]. Another example with the usage of CNNs is the work of Xu et.al. (2024) who used 3D CNNs for the converted task of binary classification to probabilistic prediction by proposing a labeling rule with soft-label for crossing period samples [24]. A multiscale STFT-based feature extraction and rectified weighting methods were combined with the 3D CNN network to capture predictive probabilities of sam-

ples and enhance the prediction [24]. These methods were tested on two datasets, namely, the CHB-MIT scalp EEG dataset and the SWEC-ETHZ iEEG dataset. According to reported results, the algorithm detected 94 out of 99 seizures during the crossing period and 100% seizures detected after EEG onset, averaged 14.84% rectified predictive ictal probability (RPIP) errors of crossing samples, 2.3 s detection latency, 0.08/h false detection rate (FDR) on the CHB-MIT dataset. The results achieved for the SWEC-ETHZ dataset were as follows: 84 out of 89 detected seizures during the crossing period, 100% detected seizures after EEG onset, 16.17% RPIP errors, 4.7 s detection latency, and 0.08/h FDR [24].

Nevertheless, working with a limited amount of data could lead to poor performance with complicated models. Thus, many existing research works implement lightweight models. For instance, Azad et.al. (2025) used a Chimera Synchronization Matrix-based Network (CSMNet) algorithm that transforms iEEG signals into 2D images with a lightweight CNN that classifies iEEG recordings into subject-independent pre-ictal, ictal, and post-ictal classes [25]. The authors used a limited data from the SWEC-ETHZ database that has two versions (short-term and long-term), reporting both accuracy scores of 96.67% on short-term iEEG recordings (excluding one patient) and 95.2% on the long-term version [25]. The mentioned chimera state is a phenomenon observed in neuronal networks where synchronized and desynchronized neuronal activities coexist, and synchronization patterns also was proven to be relevant for understanding epileptic seizures [25]. Another example is the work of Wang et. al. (2021), who used a 1D CNN network with a random selection and data augmentation (RS-DA) strategy to solve the problem of class imbalance [26]. They used two long-term datasets: the CHB-MIT scalp electroencephalogram (sEEG) dataset and the SWEC-ETHZ iEEG dataset. The used training method was event-based K-fold (K is the number of seizures per patient) cross-validation, and the evaluation was made on two levels, i.e., event-based level and segment-based level [26]. In such a way, the reported results were 88.14% sensitivity, 99.62% specificity and 99.54% accuracy in the segment-based level and 99.31% sensitivity, 0.2/h false detection rate (FDR), and a mean 8.1-s latency in event-based level for the CHB-MIT sEEG dataset.

Meanwhile, for the SWEC-ETHZ iEEG dataset, the segment-based level results were 90.09% sensitivity, 99.81% specificity, and the event-based level results were 97.52% sensitivity, 0.07/h FDR, and a mean 13.2-s latency [26].

Overall, existing seizure detection studies show that CNNs, RNNs, and transformer-based models can all be useful for iEEG analysis. However, their performance depends strongly on dataset size, patient variability, segmentation strategy, class imbalance, and the evaluation protocol. These challenges motivate the use of carefully designed windowing, class-balancing, and model-selection strategies in this thesis.

2.3 Segmentation of iEEG data

The class imbalance problem can be solved on the pre-processing level in order to increase the number of samples of ictal class. Therefore, the choice of time window that creates ictal and interictal segments plays a crucial role. In window-based segmentation, a continuous iEEG recording is divided into shorter windows of fixed duration. If the total recording duration is T , the window length is L , and the hop size between consecutive windows is H , the number of generated windows is:

$$N = \left\lfloor \frac{T - L}{H} \right\rfloor + 1 \quad (2.7)$$

where N is the total number of windows. The hop size determines how much two neighboring windows overlap. The overlap ratio can be written as:

$$\text{Overlap} = 1 - \frac{H}{L} \quad (2.8)$$

For example, Wang et.al. (2021) used the pre-processing method with a 2-s sliding window, but to increase the number of ictal segments, a varying overlap ratio was used to segment the raw ictal EEG signals during the training phase [26]. The overlap ratio was from 0.75 to 0.9, depending on the number and the time duration of seizures. For the interictal segments, the 2-s sliding windows were used without overlap [26]. Figure 2-2 shows the working principle of the sliding window method visually.

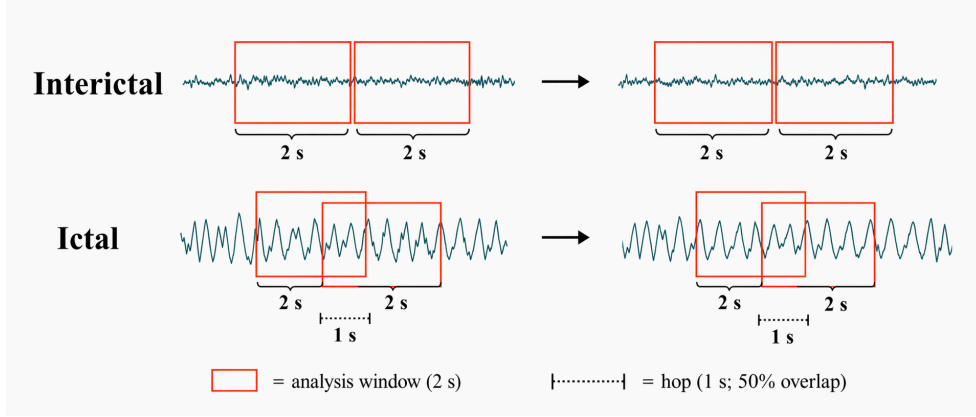


Figure 2-2: 2-s sliding windows with the overlap ratio

Another example is Sun et.al. (2025), who used 3-s samples with the variable sliding window technique, using smaller step sizes for ictal data and larger step sizes for non-ictal data to ensure an equal distribution for the training phase [23]. In such a way, authors preserved 600 samples per subject, i.e., 300 ictal and 300 nonictal samples for the SWEC ETHZ dataset. In addition, the authors worked with the HUP dataset, keeping 200 samples per subject. During the testing phase, a fixed 1-s step was employed to segment the data of each subject into 3-s samples [23].

2.4 Existing works with usage of LLMs

Review of existing solutions in terms of report generation with LLMs led to the conclusion that there is no existing solution to report generation for seizure detection using iEEG, but some systems have been developed using EEG data. One of the examples is a framework called EEGAgent developed by Zhao et.al (2025) that uses LLMs to schedule and plan multiple tools to automatically complete tasks such as information perception, spatiotemporal EEG exploration, event detection, interaction with users, and report generation [27]. The novelty of the work is the development of a general-purpose framework and does not concentrate on the analysis of EEG data on a specific task. The EEGAgent framework is built on Qwen3-235B and enhanced with an EEG-specific knowledge base. For each new task, the system embeds the task description using Qwen3-Embedding-8B and retrieves the most relevant knowl-

edge entries through similarity search [27]. It is then added to the model’s context, allowing Qwen to perform more informed EEG analysis and report generation. The core capabilities are implemented with parametric deep learning tools for capturing high-level semantics for complex tasks and non-parametric statistical and signal processing tools for extraction of fundamental features [27]. The key innovation is multi-granularity spatiotemporal analysis, with which the agent analyzes ten-second windows to detect general abnormalities. If abnormalities are found, the agent analyses the period closely in one-second windows, allowing event-level localization of epileptiform activity. The role of LLM in the system is communication between these two levels and context-based findings [27]. Regarding the report generation, evaluation of system’s expression capability was conducted using TUAB dataset aligned with the ACNS standardized template [27].

Another system that implements LLMs for reporting is called Hybrid AI, developed by Tung et. al. (2025), which concentrates only on report generation based on background EEG activity [28]. The importance of the work is the design that does not require large, annotated datasets developed for small hospitals with limited resources. The system automatically discards epochs with high amplitudes, eye movements, or unusual spectral patterns, keeping only clear segments [28]. It also implements an unsupervised outlier anomaly detection algorithm to extract 31 EEG features from segmented data, which are later used with Histogram-Based Outlier Scoring (HBOS) model to identify anomalies on the level of electrodes. So, if the electrode is determined as contaminated, the signals from neighbor electrodes are averaged [28]. Next, Hybrid AI combines several CNN architectures, namely CNN, GoogleNet, and ResNet, for posterior dominant rhythm (PDR) prediction. Obtaining other EEG features such as anterior–posterior gradients, slow-to-fast power ratios, hemispheric symmetry metrics, and focal slow-wave indices, the system generates a JSON object with key information about a patient. The JSON object is converted into text with uses Google Gemini 1.5 Pro API, the result of which is validated using few-shot prompts with OpenAI GPT-4o mini, Google Gemini 1.5 Flash, and Claude 3.5 Sonnet [28]. The reports’ content was classified as 1 or 0 based on the

presence of GBS and focal abnormality, and in case of inter-rater agreement among the three LLMs the content was evaluated using Gwet’s inter-rater reliability coefficients (AC1). The report generation accuracy resulted to be 100%, and the system significantly outperformed neurologists in detecting generalized background slowing [28].

Lastly, the usage of LLMs with brain signals was found in the work named Thought2Text, which focuses on decoding and expressing brain activity into a text using visual feature extraction and fine-tuning LLMs on image and text data [29]. However, the following work does not analyze brain signals in terms of clinical terms, which is why it is out of the scope of this literature review.

Talking about reporting templates, although structured reporting is established for scalp EEG, there is no universally standardized reporting template similar to the ACNS template for intracranial EEG (iEEG) recordings. Another example is SCORE (Standardized Computer-based Organized Reporting of EEG), which was developed through an ILAE-supported consensus process and was also intended to standardize the structure of EEG reports [30]. For invasive EEG specifically, the most relevant international guidance was the ILAE report on the diagnostic utility of invasive EEG for epilepsy surgery [31]. The document provides recommendations on indications, modalities, and techniques of iEEG. It also mentions that iEEG use varies substantially across centers, and that the recommendations are flexible for each center’s ability to design its iEEG protocols [31]. Therefore, although iEEG practice is guided by expert recommendations, there is no single universally accepted reporting format for iEEG findings across centers. A similar pattern can be seen in ACNS documents that provide technical and procedural guidance for intracranial recordings, including intraoperative electrocorticography (ECoG) and electrical stimulation with an intracranial electrode [32]. However, these documents focus primarily on minimum technical standards, documentation requirements, and procedural reporting elements rather than on a universal standardized template for iEEG reports. For example, the ACNS intraoperative ECoG guideline specifies that reports should document indication, medications, anesthesia settings, and technical recording details, and that

the technical report should separate ECoG performed to define the irritative zone from ECoG used during mapping [32]. This shows that structured documentation is recommended, but it does not amount to a single standard template for all iEEG reports.

To conclude, according to the made literature review, several findings can be determined. First, talking about HFO analysis, FRandR has been shown to be more clinically specific than ripples or fast ripples alone for epileptogenic zone localization. However, the distinction between pathological and physiological HFO is still unresolved. Second, talking about seizure detection, many studies report high performance with CNN-based, transformer-based, and hybrid models. However, results remain strongly dependent on dataset properties, segmentation strategy, and class-balancing methods. Third and most important, there is the research gap in the lack of a unified iEEG framework that combines HFO detection, seizure detection, and clinical interpretation in one system. Existing works usually address each component separately, while recent advancements in LLM-based system deal with scalp EEG data only. The use of LLMs for intracranial EEG reporting remains largely unexplored. Fourth, there is a gap in the absence of a universally accepted standardized template for iEEG reporting. Thus, this work implements a custom structured iEEG reporting template, which includes established standards in EEG reporting examples mentioned in the literature review. The template will have separate sections such as for technical context, findings, interpretation, and limitations.

Chapter 3

Methods

3.1 Comparison of old and new methodologies

Initially, the methodology had the following structure: the system performs interictal high-frequency oscillation (HFO) detection for epileptogenic zone localization as a pre-surgical diagnosis system. In addition, the ictal seizure-event detection system was considered for further development of the project. In both cases, the structured quantitative outputs would be transferred into JSON objects, which would later be transformed into a clinically formatted text report by an LLM model.

Later, the model was reconsidered to include LLM as an agentic tool instead of a report generator. The additional tools for reasoning and analysis made by LLM were developed. In this implementation, the system included two inputs for LLM agent: one for HFO detection with usage of the OpenNeuro ds003498 dataset, and a second one for seizure detection with usage of the HUP dataset. The problem with the two-input model was that no direct comparison of HFO and seizure results could be made to establish the concordance level between the results. Therefore, complex analysis was not available for one specific subject in order to provide pre or post-surgery diagnostics.

Lastly, the system was changed to a one-input system, i.e., data for one specific subject from HUP dataset was used for both seizure detection and HFO detection modules. Such implementation allows LLM to perform more complex analysis by

comparing results from both modules. In general, a new methodology includes a set of tools that are used by LLM, which has several roles to perform. Firstly, LLM is used for feature reasoning and EZ hypothesis formation given per-channel FR, R, F_{rand}R rates, spatial distribution of events, seizure onset channels, conflicting biomarkers, channel artifacts, etc. So, it forms probabilistic hypotheses about EZ/SOZ and produces channel-level likelihoods with justification. For example, it can form a hypothesis like: “Channels IAR3–4 and AHR2–3 show consistently high F_{rand}R rates and coincide with early ictal involvement in 4/5 seizures.” Secondly, LLM is reasonable for adaptive analysis with tool orchestration, i.e., to decide which tool to use at what time. So, if LLM finds the results to be ambiguous, it could rerun the reasoning stage separately. Thirdly, LLM is used for reasoning and consistency checking of information, such as detecting contradictions, flagging low-confidence conclusions, reasoning about missing electrodes, and contextualizing findings in epilepsy knowledge. Lastly, LLM is used for the report generation. Reports will be generated using a local open-source model, such as Qwen3-8B, which has advantages such as stable JSON handling and strong summarization performance. Prompts will follow a structured template. Also, the final version 4 of the LLM system was evaluated based on three criteria, such as reasoning-quality rubric, faithfulness to structured evidence, and orchestration validity. The set of tools that will be used for the system mainly consists of several blocks. The HFO detection, seizure detection, and LLM reasoning-related tools. Each of the blocks is discussed separately in the following methodology section. All of the models were trained locally with NVIDIA GeForce RTX 2060 GPU.

3.2 HFO detection

The functionality and results of HFO detection were initially tested with the OpenNeuro ds003498 dataset. The first subsection describes the whole process, describing all steps taken during the implementation. The second subsection describes how the HUP dataset was adapted for use with the existing HFO detection module. So, most

changes happened on the pre-processing stage to adjust the input type that will be suitable for the existing module. Nevertheless, the HFO detector was also changed for the HUP dataset by making it stricter.

3.2.1 OpenNeuro ds003498 dataset

The dataset used for the HFO detection task is OpenNeuro ds003498 [33]. The dataset contains HFO events recorded during NREM sleep from 20 subjects. The dataset is converted to BIDS format [33].

The following steps were implemented during dataset pre-processing:

- Take as input ieeg.vhdr, channels.tsv, and events.tsv files for each run and output ripple_epo.fif, fast_epo.fif, hfo_annotation_stats.csv
- Load events.tsv into MNE annotations
- Apply notch filter (50 Hz) and common average filter (CAR)
- Bandpass into ripples (80-250 Hz) and fast ripples (250-500 Hz)
- Make artifact rejection with RMS z-score across epochs per channel (epoch was rejected if channel exceeded threshold $|z| > 5$)
- Export annotation stats to CSV (parses annotation description strings like ripple_IAR1, fr_IAR1, frandr_IAR1, and counts them per channel with rates per minute)

Next, the algorithm was performed independently on ripple and fast ripple bands using the Hilbert-envelope thresholding method. Event-level detections were then summarized into channel-level rates and further combined into FRandR (fast ripple and ripple) co-occurrence metrics.

Epoch alignment across bands Since the artifact rejection was applied to ripple and fast ripple epochs independently, the number of epochs could differ for two files of one run. Thus, epochs were aligned using the epoch start sample index

(epoch.events[:, 0]), and only epoch whose start samples were present in both band files were kept for co-occurrence analysis.

Hilbert envelope event detection For each aligned epoch and each channel, the following steps were applied:

1. Analytic signal and envelope: the analytic signal was computed using the Hilbert transform:

$$\text{env}(t) = |\mathcal{H}\{x(t)\}| \quad (3.1)$$

2. Envelope smoothing: to reduce false threshold crossings, the envelope was smoothed using a short moving average, for example 3 ms.
3. Robust normalization: high-frequency envelopes can contain occasional outliers, so a robust z-score was computed using the median and median absolute deviation (MAD):

$$z(t) = 0.6745 \frac{\text{env}(t) - \text{median}(\text{env})}{\text{MAD}(\text{env}) + \varepsilon} \quad (3.2)$$

4. Thresholding and segment extraction: candidate HFO segments were defined as contiguous time points where $z(t) \geq z_{\text{thresh}}$, with default $z_{\text{thresh}} = 5.0$.
5. Duration constraints: candidate segments were kept only if their durations were within physiologically plausible bounds, here 6–200 ms.
6. Event characterization: for each accepted segment, the peak time was defined as the sample with maximal $z(t)$ within the segment. The detector recorded event start, end, peak time, peak robust z-score, channel name, and the epoch start sample.

The procedure was applied separately to ripple-band epoch and fast-ripple-band epochs, which resulted in ripple event and fast-ripple (FR) event tables, respectively.

FRandR co-occurrence labeling FRandR events were constructed by pairing ripple and FR events that co-occurred in the same channel within the same aligned

epoch. Pairing was performed based on event time overlap, and for each matched pair a peak timing offset was computed as follows:

$$\Delta t_{\text{ms}} = 1000 \frac{t_{\text{peak,FR}} - t_{\text{peak,ripple}}}{f_s} \quad (3.3)$$

The table generated after co-occurrence labeling includes a count of FRandR pairs per channel and a distribution of Δt_{ms} , which can be used to validate that pairings represent co-occurrence.

Channel rate computation Event counts were normalized by recording duration to compute rates per channel in events per minute. Separate rates were computed for ripple events, FR events, and FRandR pairs. For each channel c , the HFO rate was computed as:

$$\text{Rate}_c = \frac{N_c}{D} \quad (3.4)$$

where N_c is the number of detected HFO events in channel c , and D is the analyzed recording duration in minutes. These summaries form the primary structured biomarker output used for ranking channels.

HFO area percentile selection Patient or run-specific HFO areas were defined using a percentile threshold on channel rates. A channel was considered a candidate HFO-area channel if its HFO rate was higher than or equal to the 95th percentile of the channel-rate distribution:

$$c \in \text{HFO area} \quad \text{if} \quad \text{Rate}_c \geq P_{95}(\text{Rate}) \quad (3.5)$$

For a given band, such as FR or FRandR, the percentile threshold was computed across channels with positive rates. Channels with rates exceeding the threshold were labeled as `is_hfo_area = True`. This approach adapts to subject-specific baseline activity levels and yields a sparse candidate map suitable for further reasoning.

Output files For each run, the following output files were generated:

- `_hfo_detect_events.csv`: event-level detections (ripple + FR)
- `_hfo_detect_frandr_pairs.csv`: FRandR pairing table with Δt_{ms}

- `_hfo_detect_rates.csv`: per-channel rates/min for ripple, FR, FRandR
- `_hfo_detect_qc.csv`: per-epoch event counts for QC and artifact screening
- `_hfo_area_fr_95p.csv`, `_hfo_area_frandr_95p.csv`: percentile-based candidate HFO areas

3.2.2 HUP dataset

Pre-processing changes Initially, the data from the OpenNeuro ds003498 dataset was loaded as BIDS files, and for the HUP dataset, the pipeline was rewritten to read BIDS recordings stored in `_ieeg.edf` files. Also, explicit channel selection from `channels.tsv` files were added in order to remove channels marked as bad before filtering and epoching. The step is important because the recordings contain heterogeneous channel sets and may include poor-quality channels that would otherwise cause false HFO detections. The `electrodes.tsv` files were also read to include electrode coordinates to link HFO findings with anatomical channel locations, which is useful for further subject-level interpretation at the LLM stage.

For the HFO detection task, only interictal runs were selected because the developed project is intended to reflect interictal biomarker localization rather than seizure-event labeling. However, a limitation of the dataset is that some subjects have a sampling frequency of 500 or 512 Hz, while others have a sampling frequency larger than 1024 Hz. The subjects with a 500 Hz sampling frequency will have a Nyquist frequency of 250 Hz, so frequencies above 250 Hz cannot be represented properly. Therefore, in order for the FR band to be valid, the subjects selected for the HFO detection module should have a sampling frequency larger than 1000 Hz. Nevertheless, the Nyquist frequency is still close to the upper boundary of the FR band, and bandpass filtering was adjusted to 80–240 Hz for the ripple band and 250–490 Hz for the FR band.

After sorting the subjects, only 15 were found to have the sampling frequency equal to 1024 Hz: HUP126, HUP130, HUP134, HUP138, HUP139, HUP140, HUP146, HUP157, HUP160, HUP162, HUP163, HUP164, HUP165, HIP166, HUP190.

To conclude, the new pre-processing pipeline made for HUP dataset includes the following steps:

- Take as input ieeg.edf, channels.tsv, ieeg.json, and electrodes.tsv files for each run and output ripple_epo.fif, fast_epo.fif, preproc_meta.json, electrodes_kept.tsv
- Apply notch filter and common average filter (CAR)
- Band-pass filter for ripple band (80–240 Hz), and fast ripple band (250–490 Hz)
- Make artifact rejection with RMS z-score across epochs
- Epoching was made with 5-second non-overlapping windows

HFO detection changes During testing with different subjects that will be discussed in the next section, the detector was made stricter. Namely, the envelope thresholds were increased with separate thresholds for ripple and fast ripple bands, while the FR threshold was made slightly stricter than the ripple threshold, i.e., $z_threshold = 6.0$ for ripple band and $z_threshold = 6.5$ for FR band. If a channel produced too many FR events in one five-second epoch, that channel–epoch combination was flagged as suspicious and excluded from the final FR event list.

In addition, the minimum duration of candidate HFO events was increased such that the detector would make fewer brief spike-like or noise-like detections. Minimum and maximum duration for the ripple band are set to 10.0 ms and 120.0 ms, while for the FR band, the minimum and maximum duration are set to 8.0 ms and 80.0 ms.

Another addition is the requirement for a minimum number of oscillatory cycles within each candidate event. Thus, after a candidate segment was identified by the envelope threshold, the filtered waveform inside that segment was examined, and the approximate number of cycles was estimated from zero crossings. Candidate events with too few cycles were rejected. The minimum number of cycles for the ripple band is set to 4, and to 5 for FR band.

The FRandR pairing rule was also tightened. Compared to the previous implementation with simply pairing ripple and FR detections that occurred near each other

in time, the stricter version required same-channel pairing with a narrower temporal tolerance (20 ms) and actual overlap between the ripple and fast-ripple events. When several pairings were possible, the detector preferred the combination with the smallest peak-time difference and the strongest temporal overlap.

Finally, same to the previous version logic, the pipeline computes per-channel rates per minute separately for ripple, FR, and FRandR, and channels above the 95th percentile of FR rate or FRandR rate are labeled as candidate HFO-area channels. At the end, the results, such as quality-control tables, channel-wise rate tables, FR-area, and FRandR-area tables are added to JSON files for further integration into the LLM system.

3.3 Seizure detection

The seizure detection task was initially performed with the SWEC ETHZ dataset and then with HUP dataset. The results are discussed in the next section with the attempts to increase the performance.

3.3.1 SWEC ETHZ dataset

The first choice of dataset was the short-term SWEC ETHZ dataset. The short version was chosen instead of the long-term version because it would require about 1.5-2 TB of storage memory, while my laptop has limited storage space. Therefore, the choice of seizure detection model was primarily made towards lightweight and efficient networks. The dataset contains 100 anonymized iEEG datasets of 16 patients that are stored in 19 zipped folders. Each folder has several .mat files as recorded seizures (e.g., Sz1.mat, Sz2.mat, ...) for the target patient. Seizure contains preictal (3 min before seizure), ictal, and postictal segments (3 minutes before the end of recording) with expert annotations. Thus, the task of seizure detection is a three-class classification with this dataset.

The following steps were implemented during dataset pre-processing:

- Extract information from zip files
- Notch filter (50 Hz)
- Common Average Filter
- Z-score artifact rejection
- Windowing with 2-s sliding window and 1-s hop

The bandpass filtering was omitted because the dataset was already bandpass filtered (0.5-150 Hz) using a fourth-order Butterworth filter by the authors. Also, the authors applied forward and backward filtering, so the dataset already mainly had noiseless data.

For this dataset, the model combined convolutional and recurrent layers. The CNN layers were used to extract local waveform features from short iEEG windows. In this context, local features refer to short signal patterns, such as changes in amplitude or rhythmic activity, that may help distinguish between preictal, ictal, and postictal periods. The BiGRU layer was then used to model how these extracted features changed over time. A bidirectional GRU processes the sequence in both forward and backward temporal directions, allowing the model to use information from earlier and later parts of the window.

As a first attempt, three 1D CNN layers with channel attention pooling, followed by BiGRU and a fully connected head, were chosen as the model architecture. The 1D CNN networks were paired with BiGRU because hybrid models tend to perform better, since CNNs extract local features, and GRU captures temporal dependencies. In addition, GRU was used instead of LSTM because it is simpler and faster to train. Channel attention pooling was added to give higher weight to more informative channels and lower weight to less informative or noisier channels.

At the end of the model, a fully connected layer produced one score for each of the three classes. These scores were converted into class probabilities using the softmax function:

$$p_i = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}} \quad (3.6)$$

where p_i is the predicted probability for class i , z_i is the model output score before normalization, and C is the total number of classes. In this case, $C = 3$, corresponding to the preictal, ictal, and postictal classes. The final predicted class was selected as the class with the highest probability.

The exact configuration of CNN layers was kernel sizes of 9, 5, and 5 with a stride equal to 2 (temporal downsampling). The next layer is channel attention over the channels layer, which uses weighted aggregation that decides which channel should contribute more. Some channels could be noisier, while others can be closer to the generators of seizures. The hidden size of BiGRU was 128. At the end, a fully connected layer made an output using a 3-class softmax classification (pre-ictal, ictal, post-ictal). The training was conducted with a batch size of 64, dropout 0.3, 50 epochs, AdamW optimizer, balanced sampling (WeightedRandomSampler), and focal loss ($\gamma = 2.0$ with class weights) to deal with class imbalance. The focal loss is better for the imbalance class case because non-ictal windows contribute less, compared to ictal windows.

3.3.2 HUP dataset

HUP dataset contains iEEG recordings from 58 subjects [34]. The data was converted into BIDS format.

The following steps were implemented during dataset pre-processing:

- Notch filter (60, 120 Hz)
- Bandpass filter (0.5 - 150 Hz)
- Common Average Filter
- Z-score artifact rejection
- Windowing with 2-s sliding window and variable hop (nonictal hop = 1 s, ictal hop = 0.25 s)

For the HUP dataset, the seizure detection task was treated as binary classification, i.e., ictal vs non-ictal. In other words, each window was assigned either to the

seizure class or to the non-seizure class. This formulation was used because the main goal was to detect ictal seizure activity in the HUP recordings.

Since the dataset is much larger than SWECH ETHZ, the chosen model included an attention mechanism. A 1D CNN encoder was applied independently to each channel to extract local temporal features from the iEEG signal. The resulting channel-level feature vectors were then passed to a transformer encoder applied across channels. In this architecture, the transformer helps the model learn relationships between electrodes and identify which channels are more informative for seizure detection. The final binary classification was based on a CLS token. The CLS token is a learnable summary representation that collects information from all channel features and is then used by the classifier to predict whether the window is ictal or non-ictal.

For binary classification, the model produced one output score z , which was converted into a seizure probability using the sigmoid function:

$$p = \sigma(z) = \frac{1}{1 + e^{-z}} \quad (3.7)$$

where p is the predicted probability that the input window belongs to the ictal class. The final class label was assigned using a decision threshold τ :

$$\hat{y} = \begin{cases} 1, & p \geq \tau \\ 0, & p < \tau \end{cases} \quad (3.8)$$

where $\hat{y} = 1$ indicates an ictal prediction, $\hat{y} = 0$ indicates a nonictal prediction, and τ is the decision threshold. For example, when $\tau = 0.5$, a window is classified as ictal if the predicted seizure probability is at least 0.5. In later experiments, threshold tuning was used to adjust this value and improve the balance between sensitivity and false-positive predictions.

As the first attempt, the CNN network had 64 hidden channels and 128 embedding dimensions. Meanwhile, the number of transformer layers was 2, and the number of transformer heads was 4. The training procedure included leave-one-run-out validation, a fixed decision threshold equal to 0.5, a batch size of 32, dropout 0.2, 35

epochs, early stopping equal to 8, a weighted random sampler, and binary focal loss.

3.4 LLM

A large language model (LLM) is a neural network trained to model and generate text by predicting the next token in a sequence. Modern LLMs are commonly based on transformer architectures, which use attention mechanisms to model relationships between tokens in a text sequence [22]. In general, the probability of a generated sequence can be written as:

$$P(x_1, x_2, \dots, x_n) = \prod_{t=1}^n P(x_t \mid x_1, \dots, x_{t-1}) \quad (3.9)$$

where x_t is the token generated at step t , and the previous tokens $x_1, \dots, x_{t-1}$ provide the context for predicting the next token. This means that the model generates text one token at a time, using the preceding tokens as context.

In this thesis, the LLM was not used to analyze raw iEEG signals directly. Instead, it received structured summaries generated by the HFO detection and seizure detection modules. These summaries were stored in JSON format and included information such as HFO candidate channels, FR and FRandR rates, seizure probability segments, candidate ictal intervals, channel-level seizure evidence, and cross-modal overlap. The role of the LLM was to reason over these structured findings and produce a human-readable report.

JSON was used as an intermediate representation because it stores results in a structured and machine-readable format. This reduced ambiguity and allowed the LLM to reason over predefined fields rather than unstructured text. In addition, the use of JSON made it possible to apply schema validation, consistency checks, and deterministic report rendering in later versions of the system. Therefore, the LLM acted as a reasoning and reporting component operating on structured evidence, rather than as a model that directly classified raw iEEG recordings.

3.4.1 Version 1

To begin, the main purpose of this version was the development of a unified system with input from one specific subject from the HUP dataset, where the LLM model is used to create a structured report. The version 1 system was developed with the Qwen3:8B model. The following directory structure was created for the system. The exact folder tree view can be seen in Listing 3.1:

Listing 3.1: Directory organization of the Version 1 LLM pipeline.

```
LLM_unified_HUP/  
|-- prompts/  
|   |-- reasoning_prompt.txt  
|   |-- report_prompt.txt  
|-- sub-HUP146/  
|   |-- hfo/  
|       |-- sub-HUP146_hfo_summary.json  
|   |-- seizure/  
|       |-- sub-HUP146_ictal_run_summary.json  
|       |-- sub-HUP146_interictal_run_summary.json  
|   |-- integration/  
|       |-- sub-HUP146_master_case.json  
|   |-- reports/  
|       |-- sub-HUP146_compact_llm_input.json  
|       |-- sub-HUP146_reasoning.json  
|       |-- sub-HUP146_reasoning_raw.txt  
|       |-- sub-HUP146_report_raw.txt  
|       |-- sub-HUP146_final_report.txt  
|-- build_master_case.py  
|-- run_llm_case.py
```

The prompts folder contains two prompt .txt files: reasoning prompt and report prompt. In general, the system uses a two-stage LLM pipeline. Firstly, the LLM uses a reasoning prompt and a compact case JSON that is reduced input with summary from both branches as inpput and outputs a reasoning JSON file stored in the results

folder. The reasoning JSON has fields such as case assessment, HFO assessment, seizure assessment, integration assessment, limitations, and final interpretation generated by the LLM model. Secondly, LLM uses a report prompt, the compact case JSON, and the reasoning JSON from stage 1 to create a final report in txt format. Its role was to transform the structured reasoning into a final technical report with fixed sections, such as system summary, HFO module summary, seizure module summary, integration constraints, overall interpretation, and limitations. This separation into two systems improved the debugging process, so if the final report was poor, it could have been determined whether the issue originated in reasoning or only in text generation.

So, in order for LLM to output a structured report with reasonable findings prompts in version 1 were designed to be restrictive. They instructed the model to use only information explicitly present in the JSON inputs, avoid unsupported inferences, avoid inventing seizure onset channels or spatial overlap, state uncertainty clearly, and distinguish observed findings from interpretation. Later, the requirement of explicit distinction between strong-consensus HFO channels and moderate candidate HFO channels was added, because reports generated by LLM mentioned all candidate HFO channels as strong-consensus channels.

The next folder is the sub-HUP_146 folder, which contains all JSON files resulting from seizure and HFO detection in seizure and hfo folders respectively, a master JSON file that combines information from both branches in the integration folder, and all output files generated by LLM model in reports folder.

So, regarding each folder separately, the seizure folder has two JSON files stored inside: one is a summary of the ictal run, and another is a summary of the interictal run that was held out for LLM testing. These JSON files have information about window-level counts, predicted-positive fraction, smoothed probabilities, threshold used, candidate positive segments, and per-window predictions extracted after seizure detection branch. The hfo folder contains only one JSON file with a summary of the HFO detection branch with information about ripple and FR detections, FRandR pairings, channel-level event rates, HFO-area summaries, and cross-run aggregated

evidence. The integration folder contains a master JSON file that is generated by the `build_master_case.py` code, which is the most important development stage for this version. The master JSON combines the summaries of two branches, and stores the subject ID and dataset metadata, summaries of three input JSON files, integration-level constraints, and task instructions for the LLM.

The reports folder stores all output files generated by running `run_llm_case.py` code. The input to `run_llm_case.py` initially was the master JSON, but such an implementation led to unstable outputs. In such a way, a compact case JSON that was mentioned earlier was created to reduce the input content to the LLM model, which is also stored in the results folder. Thus, the system outputs compact case JSON, reasoning JSON, the final report in txt format, as well as raw, reasoning, and raw report txt files that are intermediate results from stage 1 and stage 2, respectively.

3.4.2 Version 2

The main goal of this version was to improve the integration of HFO detection and seizure detection outputs by adding interpretable seizure-channel evidence, deterministic cross-modal fusion logic, and consistency checking before report generation. Previously, LLM used mainly information from summary JSONs without much reasoning between the two branches. Although the HFO branch already produced channel-level evidence, the seizure branch mainly described ictal probability trajectories and candidate ictal segments, without directly exposing which channels contributed most strongly to the classifier’s decision. As a result, cross-modal comparison between HFO channels and seizure-related channels was limited.

Among the made changes, firstly, the seizure detection module was changed to export channel-level seizure evidence, which was obtained using a $\text{gradient} \times \text{input}$ saliency proxy. This means that the model’s output for ictal prediction was analyzed to estimate which channels contributed most strongly to the classifier’s decision during early ictal windows. Also, the saliency proxy is used only as an approximation, and the channels were not interpreted as true electrophysiological onset channels. In the compact case JSON, this is explicitly recorded as `gradient_x_input_saliency_proxy`,

and LLM was constrained to state that seizure channel-level evidence is a saliency proxy rather than direct onset annotation. Also, some subjects were selected for LLM testing, and out of all runs for a specific subject, one ictal run and one interictal run were reserved exclusively for inference and LLM evaluation. These two runs were excluded from seizure-model training and validation and were used only after model fitting to generate JSON summaries for the LLM pipeline. However, the remaining runs from the same subject were used in the training set. Therefore, the LLM case studies represent a run-level held-out evaluation within the subject, rather than a fully subject-independent evaluation. For all subjects, regardless of whether they were held out for LLM testing, both ictal and interictal runs were used during the training process.

Secondly, this version introduces the replacement of anonymous model-channel indices, such as CH058 with real bipolar electrode names such as RG1-RG2, RG3-RG4, and RH1-RH2. The mapping was created from electrodes.tsv files, which are individual for each subject. The mapping was added to export the JSON file with other seizure detection results. For example, Table 3.1 shows the example of mapping for subject HUP146 for a few channels. This step substantially improved interpretability because seizure saliency outputs could now be compared directly with HFO candidate channels at the electrode-pair level.

Table 3.1: Mapping between model channel identifiers and bipolar electrode names for the seizure-detection branch of subject HUP146.

Model channel	Electrode pair
CH001	LG1-LG2
CH002	LG2-LG3
CH003	LG3-LG4
CH004	LG4-LG5
CH005	LG5-LG6
CH006	LG6-LG7
CH007	LG7-LG8
CH008	LH1-LH2
CH009	LH2-LH3
CH010	LH3-LH4

Thirdly, this version adds deterministic cross-modal integration and consistency

checking. It means that LLM does not operate with a raw signal, but deterministic tool written in Python scripts handles the task. In this version, the `integration_logic.py` script was used to determine channel concordance level and perform consistency checks. The concordance level section is used for comparing of HFO candidate channels against seizure saliency-proxy channels and reporting HFO candidate channels, seizure candidate channels, overlap channels, overlap count, concordance level, and a short textual summary. The consistency checking evaluated whether the HFO and seizure branches were supportive or contradictory to each other.

Lastly, the compact case JSON was changed in structure and added more interpretation constraints. The compact case JSON now has the following sections: system-level view, HFO candidate channels, seizure run summaries, seizure channel-level evidence, deterministic channel concordance, deterministic consistency checks, integration constraints, and LLM constraints. The LLM model was constrained such that seizure channel evidence must be described as supportive model-derived evidence, direct HFO–seizure overlap should not be interpreted as definitive localization, disagreements between modalities should lower confidence rather than be forced into a single conclusion, and no unsupported diagnoses, onset locations, or outcomes should be invented.

Except for the addition of `integration_logic.py`, the rest of the directory remained unchanged compared to version 1.

3.4.3 Version 3

The main goal of this version was to improve the reliability, controllability, and reproducibility of the multimodal integration stage after the cross-modal fusion logic introduced in version 2. The previous version’s LLM stage was still relatively fragile because it produced shallow or inconsistent reasoning that will be discussed in the next section in more detail. So, the changes made are related to the improvement of the final reasoning and reporting stages that were not yet sufficiently stable.

Firstly and most importantly, this version adds a strict reasoning JSON schema together with explicit validation and repair. In the previous version, the LLM could

still return outputs that had reasonable findings but were poorly structured, too generic, or inconsistent with the evidence in the compact case JSON. In version 3, the LLM was constrained to produce a fixed reasoning JSON with predefined sections such as overall confidence, confidence rationale, HFO assessment, seizure assessment, integration assessment, limitations, and EZ hypothesis. After generation, the reasoning output is validated by Python code, and if the output does not satisfy the required schema or content constraints, a repair prompt is used to correct it. This addition significantly improved the reliability of the reasoning stage.

Secondly, this version extends the constraints imposed on the LLM further. For example, the confidence rationale had to mention specific evidence factors, ictal summaries had to include timing information when candidate segments were present, limitations had to be explicitly provided under mismatch or proxy-evidence conditions, and cross-modal mismatch summaries had to mention their effect on confidence. Such changes introduced evidence-grounding to the reasoning of LLM.

Thirdly, this version standardizes final report generation into fixed sections. In version 2, the final LLM output could still vary in structure and phrasing, which made it harder to guarantee consistent output quality. In version 3, the final report was rendered by Python from the validated reasoning JSON and the compact case JSON. The report was generated using stable predefined sections such as case overview, HFO findings, seizure findings, cross-modal integration, limitations and confidence, and final impression. This change improved reproducibility and made the final system output more predictable.

Lastly, version 3 reorganized the system and has a different directory. Compared to version 2, separate folders and scripts were introduced for integration, LLM reasoning, reporting, and utility functions. As a result, the workflow became easier to follow: branch summaries are first fused deterministically, then reduced into a compact LLM-ready case representation, then interpreted by the LLM under a validated schema, and finally rendered into a fixed report format. Therefore, in contrast to version 2, which mainly improved the content of multimodal integration, version 3 mainly improved the architecture and reliability of multimodal reasoning and report generation.

3.4.4 Version 4

The goal of the final version was to extend the system where LLM is used as a reasoner to a pipeline with LLM performing tool orchestration, meaning that it was also given a planning role, deciding which system functions should be executed next based on the current compact case. For this purpose, a new planner schema and planner prompt were introduced. The new LLM system has the following directory, and the full tree view can be seen in Listing 3.2:

Listing 3.2: Directory organization of the Version 4 LLM system.

```
LLM_unified_HUP_ver4/
|-- config/
|   |-- reasoning_schema.json
|   |-- planner_schema.json
|-- prompts/
|   |-- reasoning_prompt.txt
|   |-- repair_reasoning_prompt.txt
|   |-- planner_prompt.txt
|   |-- repair_planner_prompt.txt
|-- integration/
|   |-- __init__.py
|   |-- compact_case_builder.py
|   |-- fusion_logic.py
|-- llm/
|   |-- __init__.py
|   |-- ollama_client.py
|   |-- reasoning_runner.py
|   |-- reasoning_validator.py
|-- tools/
|   |-- __init__.py
|   |-- tool_registry.py
|   |-- tool_executor.py
|   |-- integration_tools.py
|-- orchestration/
```

```

|   |-- __init__.py
|   |-- orchestration_runner.py
|   |-- planner_runner.py
|   |-- planner_validator.py
|-- reporting/
|   |-- __init__.py
|   |-- report_renderer.py
|-- sub-HUP146/
|   |-- hfo/
|   |   |-- sub-HUP146_hfo_summary.json
|   |-- seizure/
|   |   |-- sub-HUP146_ictal_run_summary.json
|   |   |-- sub-HUP146_interictal_run_summary.json
|   |-- integration/
|   |   |-- sub-HUP146_master_case.json
|   |   |-- sub-HUP146_compact_case.json
|   |-- reasoning/
|   |   |-- sub-HUP146_reasoning_raw.txt
|   |   |-- sub-HUP146_reasoning_repair_raw.txt
|   |   |-- sub-HUP146_reasoning_validation_reprot.json
|   |   |-- sub-HUP146_reasoning_validated.json
|   |-- orchestration/
|   |   |-- sub-HUP146_orchestration_state.json
|   |   |-- sub-HUP146_tool_execution_log.json
|   |   |-- sub-HUP146_tool_plan_raw.txt
|   |   |-- sub-HUP146_tool_plan_validated.json
|   |   |-- sub-HUP146_tool_plan_validation_report.json
|   |-- reports/
|   |   |-- sub-HUP146_final_report.txt
|   |   |-- sub-HUP146_report_data.json
|-- run_orchestration.py
|-- run_reasoning.py
|-- run_report.py
|-- run_pipeline.py

```

The LLM performing the new rule as planner had to generate a strict tool plan

JSON that contained the following sections: case identifier, subject identifier, planning decision, planner reason, and ordered action list. The action list consisted of named tools and step numbers. Only a restricted set of tool names was allowed. Thus, the planner output represented a bounded action plan rather than open-ended agent behavior.

Similar to version 3, planner output was validated before being executed. The planner validator checked the presence of required keys, validity of planning decision, validity of tool names, correct action ordering, and consistency between the action list and planning decision. This step ensures that LLM cannot use invented tools. In case the planner output was recognized as invalid, a repair prompt was used. Only validated plans were allowed to proceed to execution.

The newly added tool orchestration layer uses several Python scripts. Such a structure separates planning, validation, execution, and logging into separate blocks. The function of each script is described in Table 3.2.

Table 3.2: Main software components used in the Version 4 orchestration framework.

Component	Function
<code>tool_registry.py</code>	Maps tool names to executable Python functions.
<code>tool_executor.py</code>	Executes the validated tool plan step by step and logs the result of each tool call.
<code>integration_tools.py</code>	Contains selected internal functions exposed as tools that can be called by the planner.
<code>orchestration_runner.py</code>	Connects the planner, validator, and executor into a single bounded orchestration loop.
<code>run_orchestration.py</code>	Provides a terminal entry point for running orchestration on a subject.

The first tools introduced for version 4 are represented in Table 3.3. The planner uses these tools, looking at the current compact case. For example, if the compact case has `concordance_level = none`, the `cross_modal_mismatch` indicating consistency flag, the system may decide to rebuild the integrated case, regenerate the compact case, rerun reasoning, and regenerate the report.

The complete workflow of version 4 can be summarized as follows. Figure 3-1 represents the full workflow of version 4 as flowchart:

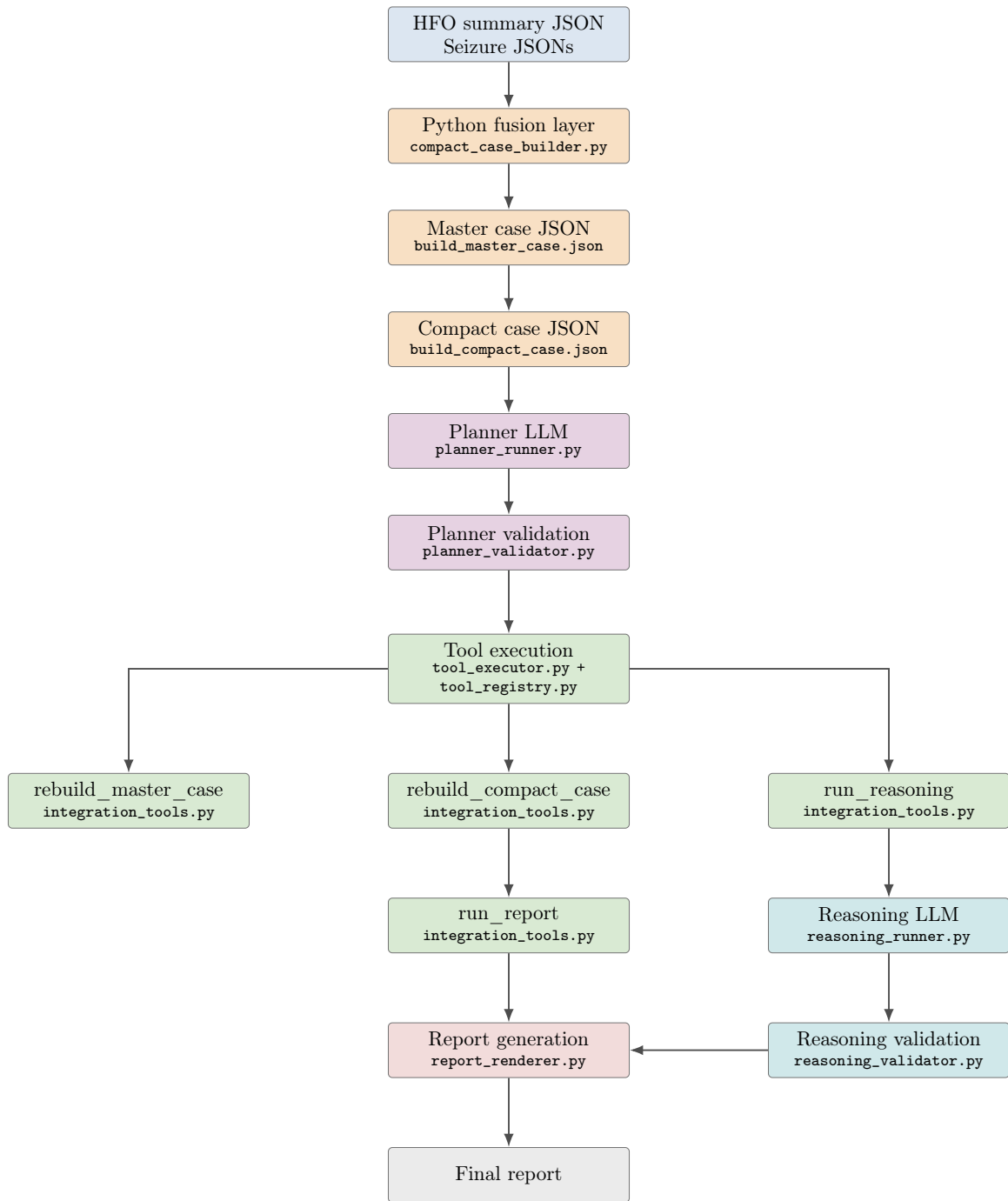


Figure 3-1: Architecture of the Version 4 LLM system with planner-based tool orchestration.

Table 3.3: Tools available to the Version 4 planner.

Tool	Purpose
<code>rebuild_master_case</code>	Reconstructs the full multimodal case from HFO and seizure outputs.
<code>rebuild_compact_case</code>	Produces a reduced LLM-ready representation of the case.
<code>run_reasoning</code>	Runs schema-constrained LLM reasoning and validation.
<code>run_report</code>	Regenerates the final structured report.

1. Load current `compact_case.json`
2. Planner LLM reads the compact case
3. Planner LLM outputs a strict tool plan
4. Python validates the tool plan
5. Python executes tools sequentially
6. The case state is refreshed
7. Reasoning and report artifacts are updated

3.4.5 Rubric Evaluation

The system evaluation was decided to be made for the final version of the system. The evaluation was made for the following subjects: sub-HUP126, sub-HUP130, sub-HUP146, sub-HUP157, and sub-HUP164. The subjects were chosen randomly across the pool of 15 subjects with a sampling frequency of 1024 Hz to avoid problems on the HFO detection stage. The Results section will describe the development process in details for subject sub-HUP146, while final reports of other subjects can be found in the Appendix A section. Since no formal expert review was available, the evaluation was designed as a technical assessment of the system’s reliability, faithfulness to structured multimodal evidence, reasoning quality, and bounded tool orchestration behavior. For each subject, the final system was run end-to-end, producing planner outputs, validated reasoning JSON, final reports, and orchestration logs.

Rubric-based reasoning quality The Table 3.4 represents the rubric used for the reasoning quality of the LLM system. The maximum number of given points can be equal to 10. 0 = poor/absent 1 = partial/acceptable 2 = good/complete

Table 3.4: Rubric used for evaluating the reasoning quality of the final LLM system. Scores are assigned as 0 = poor/absent, 1 = partial/acceptable, and 2 = good/complete.

Criterion	0	1	2
Evidence coverage	Major findings from one or both branches are missing	Some key findings are included, but others are omitted	Key HFO, seizure, and integration findings are all included
Cross-modal reasoning	Modalities are listed separately without comparison	Some comparison is present, but weak or incomplete	Explicit comparison between HFO and seizure branches is provided
Uncertainty handling	Output is overconfident or ignores mismatch/limitations	Some uncertainty is mentioned, but incompletely	Confidence is adjusted appropriately and limitations are clearly stated
Constraint adherence	Unsupported claims, invented channels, or misuse of evidence are present	Mostly constrained, but minor issues remain	Fully respects constraints, including proxy wording and no unsupported claims
Clarity and structure	Difficult to follow or poorly organized	Understandable but uneven or repetitive	Clear, well-structured, and easy to interpret

Faithfulness to structured evidence The Table 3.5 represents a checklist that measures whether the output is faithful to the input evidence.

Evaluation of tool orchestration The Table 3.6 shows a checklist that evaluates LLM in the role of planner.

Table 3.5: Faithfulness checklist used to verify whether the generated reasoning and report remain consistent with the structured compact case JSON.

Item	Yes/No
HFO strong consensus channels are reported correctly	
HFO supportive channels are reported correctly	
Seizure supportive channels are reported correctly	
Ictal segment timing is preserved when present	
Concordance level matches compact case	
Overlap / no-overlap status matches compact case	
Seizure evidence is correctly described as model-derived saliency proxy	
Confidence direction is consistent with mismatch / consistency flags	
No invented channels are present	
No unsupported diagnosis, onset location, or outcome is introduced	

Table 3.6: Tool orchestration checklist used for evaluating the bounded planner and execution behavior of version 4.

Item	Yes/No
Planner output is valid JSON	
Planning decision is appropriate for the case state	
Only allowed tool names are used	
Action order is valid	
Tool execution completes successfully	
Updated reasoning JSON is valid	
Final report is regenerated successfully	

Chapter 4

Results and Discussion

4.1 HFO detection

4.1.1 OpenNeuro dataset

Pre-processing results The result of the OpenNeuro dataset pre-processing can be seen in Figure 4-1, where Figure 4-1a represents time series results for raw signal, signal after notch filter and signal after notch filter with CAR (Common Average Reference) and Figure 4-1b represents PSD plot for raw signal, signal after Notch filter and signal after Notch filter with CAR.

For the Figure 4-1a, the green graph showing the Notch filter effect with CAR is less noisy compared to the orange graph that shows only the Notch filter effect, as it should be because CAR averages the signal. For the Figure 4-1b, the Notch filter was applied for 50 Hz and its harmonics, i.e., 50 Hz, 100 Hz, 150 Hz, ..., 550 Hz, skipping filtering around 250 and 500 Hz to preserve the boundaries of ripple and fast ripple bands. The result can be clearly seen from the peaks near the harmonic frequencies from the orange graph of 4-1b.

HFO detection results The result of the HFO detection algorithm can be analyzed using the output .csv files for a single subject (sub-01, interictal sleep, run-01). Firstly, looking into Table 4.1, several conclusions can be made. The FR ranking shows that the FR rates are highly concentrated, i.e., the ranking shows one dom-

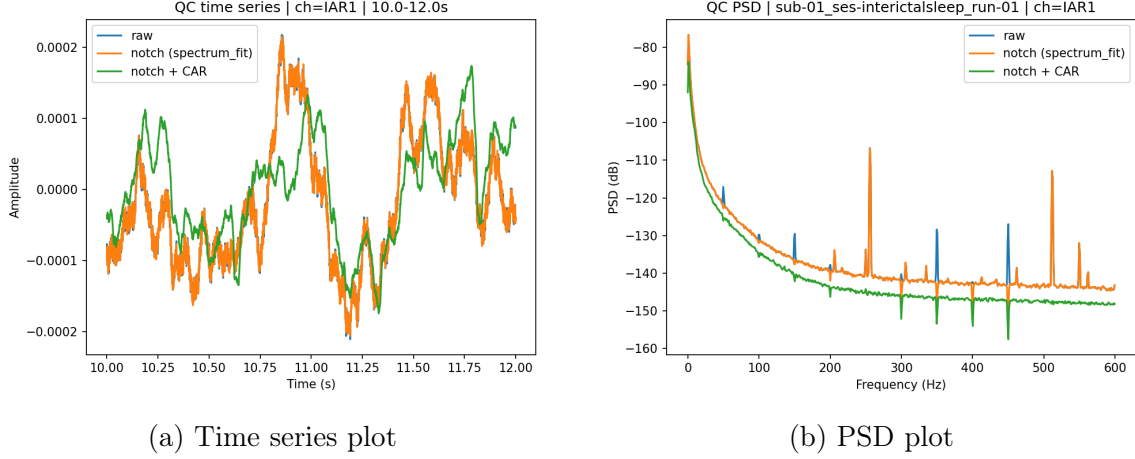


Figure 4-1: Results of pre-processing of OpenNeuro dataset

inant channel (PHR1) with approximately 20.4 events/min, followed by a second channel (PHR2) at 10.5 events/min, and then a drop to a smaller cluster around 6.5 events/min. This pattern indicates that FR activity was not uniformly distributed but instead concentrated on a small subset of contacts. For the FRandR ranking, the similarity with the FR ranking can be seen with two dominant channels being PHR1 and PHR2. This overlap between top FR and top FRandR channels strengthens the interpretation that the dominant channels are consistent with a high-frequency burden across related biomarkers rather than isolated detections. In contrast, the ranking of ripple band shows that ripple rates are more distributed, i.e., have a broader set of channels and magnitude values larger than FR/FRandR. These observations prove that ripples are a more frequent phenomenon, which can include both pathological and physiological events. As a result, ripples alone are expected to be less specific for localization than FR or FRandR.

The percentile-based HFO area heuristic (95th percentile) selected a very small channel subset in this run, which is shown in the Table 4.2. Specifically, both the FR and FRandR HFO areas contained only two channels (PHR1 and PHR2), with thresholds near 9.5–10.1 events/min. Such a sparse selection is desirable for precise localization purposes.

Per-epoch QC plot in Figure 4-2 shows high ripple counts and comparatively low FR counts, as expected. The ripple plot has some peaks, which did not cause the

Table 4.1: Top 15 channels ranked by FR, FRandR, and ripple rates.

Rank	FR			FRandR			Ripple		
	Channel	Count	Rate/min	Channel	Count	Rate/min	Channel	Count	Rate/min
1	PHR1	97	20.421	PHR1	97	20.421	AR2	354	74.526
2	PHR2	50	10.526	PHR2	51	10.737	AR1	250	52.632
3	AHR2	31	6.526	PHR8	36	7.579	PHR1	238	50.105
4	PHH8	31	6.526	AHR1	31	6.526	AHR2	229	48.211
5	AR2	31	6.526	AR2	30	6.316	AR3	194	40.842
6	AHR1	30	6.316	AHR2	29	6.105	AHR6	188	39.579
7	PHR7	14	2.947	PHR7	15	3.158	AHR1	184	38.737
8	AR1	12	2.526	AR1	12	2.526	AR4	182	38.316
9	AR3	9	1.895	AR3	9	1.895	AHR8	180	37.895
10	PHR3	7	1.474	PHR3	7	1.474	PHR2	173	36.421
11	PHR6	7	1.474	PHR6	7	1.474	AHR7	161	33.895
12	AHR3	6	1.263	AHR3	7	1.474	AHR3	151	31.789
13	AR4	5	1.053	AR4	6	1.263	AHR5	137	28.842
14	LAR5	3	0.632	PHR5	2	0.421	IL1	137	28.842
15	PHR5	2	0.421	IAR5	2	0.421	AR8	120	25.263

Table 4.2: HFO area determined by 95th percentile.

(a) FR area 95p: threshold=9.5263, selected_channels=2/26

	channel	band	count	rate_per_min	threshold	is_hfo_area
18	PHR1	fr	97	20.421053	9.526316	True
19	PHR2	fr	50	10.526316	9.526316	True

(b) FRandR area 95p: threshold=10.1053, selected_channels=2/25

	channel	band	count	rate_per_min	threshold	is_hfo_area
17	PHR1	frandr	97	20.421053	10.105263	True
18	PHR2	frandr	51	10.736842	10.105263	True

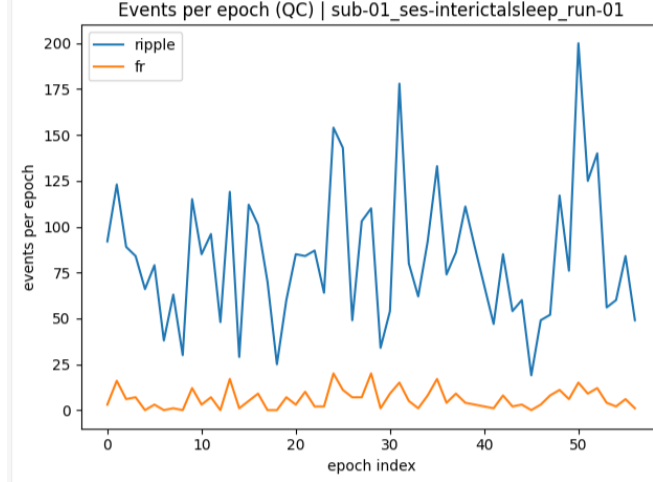


Figure 4-2: Events per epoch for sub01, run01

same peaks in the FR plots. Thus, it also argues with the observation made before. Nevertheless, epochs with extreme ripple event counts remain useful flags for further review or for applying stricter artifact criteria.

The FRandR co-occurrence timing offsets (Δt_{ms}) in Figure 4-3 show a strong central clustering near 0 ms, with a median close to 0 and an interquartile range within a few milliseconds. This pattern supports that the pairing procedure is capturing temporal co-occurrence between ripple and FR features within the same channel. The presence of long tails (tens of milliseconds) suggests that the co-occurrence tolerance window may be permissive.

Finally, it is important to conduct cross-run and cross-subject analysis, which are highly meaningful for evaluating the detector’s qualitative behavior and robustness. Table 4.3 shows how focal the biomarker distribution was within each run, which was quantified with two concentration measures (top-5 share and Gini coefficient) computed from per-channel rates. The top-5 share demonstrates the fraction of the total rate contained in the top 5 channels, and the Gini coefficient is used as an inequality measure of rate distribution across channels. Comparing Tables 4.3a and 4.3b, it can be seen that FR and FRandR distributions were strongly concentrated (Gini 0.67 and top-5 share 0.69–0.70), while ripple distributions were substantially less concentrated (Gini 0.43 and top-5 share 0.27). This supports the interpreta-

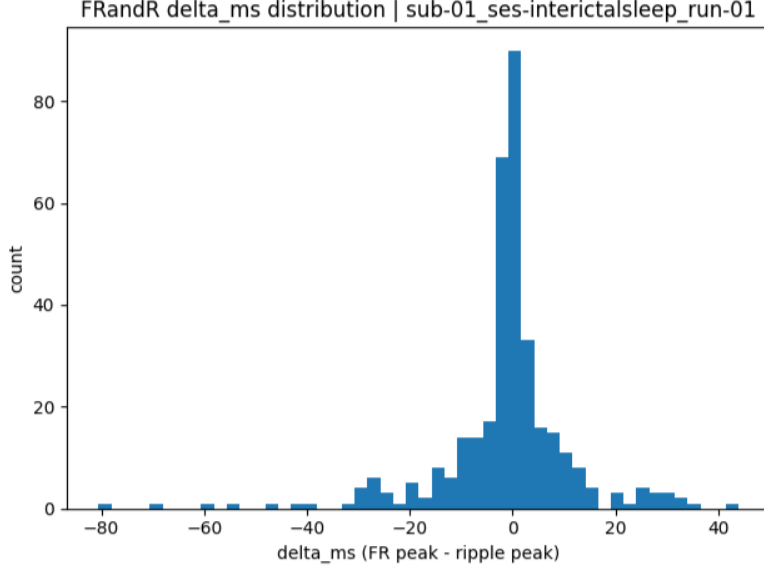


Figure 4-3: FRandR co-occurrence shown in Δt_{ms} distribution

tion that higher-frequency biomarkers (FR/FRandR) are intrinsically more focal and potentially more localizable than ripples.

Table 4.3: All runs summary with top-5 share and Gini coefficient.

(a) FR summary					(b) FRandR summary				
Run	Events	FR rate	Top-5	Gini	Run	Pairs	FRandR rate	Top-5	Gini
run-01	5085	73.263	0.690	0.675	run-01	352	74.105	0.696	0.671
run-02	4122	58.105	0.659	0.693	run-02	238	50.105	0.777	0.729
run-03	4037	52.071	0.823	0.684	run-03	250	53.571	0.836	0.650
run-04	3958	58.929	0.735	0.711	run-04	255	54.643	0.784	0.678
run-05	3267	38.316	0.912	0.663	run-05	181	38.105	0.912	0.635

Across all runs, the histogram of FRandR Gini values shown in Figure 4-4 summarizes the dataset-level tendency of the detector to produce focal maps. A distribution peaking at moderately high Gini values indicates that, for most runs, FRandR activity is concentrated within a small subset of channels rather than being uniformly distributed. Runs with lower Gini values correspond to more diffuse distributions, potentially reflecting widespread interictal activity, differences in sleep state, or residual artifacts.

Therefore, the following main findings can be summarized from HFO detection results:

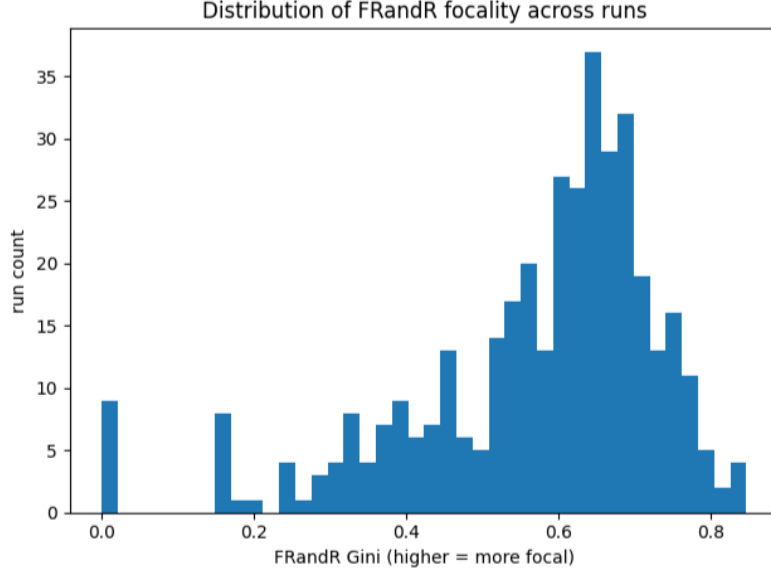


Figure 4-4: Histogram of FRandR Gini values for all runs

- FR and FRandR rates are significantly more focal than ripple rates, which is consistent with their expected higher specificity.
- HFO area selection ended with two dominant channels in the example run, which is appropriate for localization.
- FRandR co-occurrence timing clustered tightly around 0 ms, supporting that the detected co-occurrence reflects true temporal coupling rather than random pairing.
- Across 382 runs, concentration metrics (Gini/top-share) provide a scalable way to characterize detector behavior.

4.1.2 HUP dataset

As mentioned before, only 15 subjects were determined to have the sampling frequency equal to 1024 Hz. The first subject selected for the HUP dataset pre-processing was subject HUP164, but later the pre-processing was also run on subject HUP146 for further testing with the LLM system.

So, interictal run01 and run02 of subject HUP164 were tested with the initial pre-

processing pipeline that remained unchanged from the version with the OpenNeuro dataset. Such a configuration led to the following results:

Run-01 has 43 epochs, 6102 ripple events, 3079 FR events, and 2521 FRandR pairs. Its FR/FRandR activity was dominated by channels such as LA12, RA11, and RA12 with extremely high rates. Also, the 95th-percentile FR area was found to be LA12, RA10, RA11, RA12, and the 95th-percentile FRandR area was LA12, RA11, RA12.

Run-02 has 29 epochs, but only 739 ripple events, 39 FR events, and 21 FRandR pairs. Its FR and FRandR rates are much lower compared to run01. The FR 95th-percentile area was only RA12, while the FRandR 95th-percentile area was LC2, LH11, LH12, RA11, RA12.

One of the most concerning findings was the difference in FR events between runs. In detail, in run-01, the count of FR events was 3079, and in run-02, the count was 39. Such a drastic difference between runs shows that run01 has high pathological high-frequency activity and that the detector is probably too permissive on that run. In addition, the overlap across runs was partial, i.e., some channels appear in both runs (RA11, RA12, LC2, LH11, LH12), which is a good sign. However, the strongest FR and FRandR channel in run01 was LA12, which was barely present in run02.

Thus, the detector was changed as it was described in the HFO detection changes part of the Methods section. New results were much better for run01 with 43 epochs, 130 ripple events, 31 FR events, and 19 FRandR pairs. The run02 inspection showed little FR activity: 29 epochs, 6 ripple events, 3 FR events, and no FRandR pairs.

The most important improvement is cross-run recurrence. So, in run-01, top FR channels include LC1, RB3, LC2, RA12, LC3, RA11, and top FRandR channels include RB3, LC1, LC3, LC2, RA4. The FR 95th-percentile area is LC1, and the FRandR 95th-percentile area is RB3. In run02, the FR channels are LH11, LH12, and RA11, and the only FRandR channel is RA11. The FR area is LH11, LH12, and RA11, and the FRandR area is also RA11. So, RA11 is your strongest stable candidate, because it appears in both runs. The less stable candidates appeared to be LC1, RB3, LC2, and LC3 because they appear in run-01, but do not replicate in

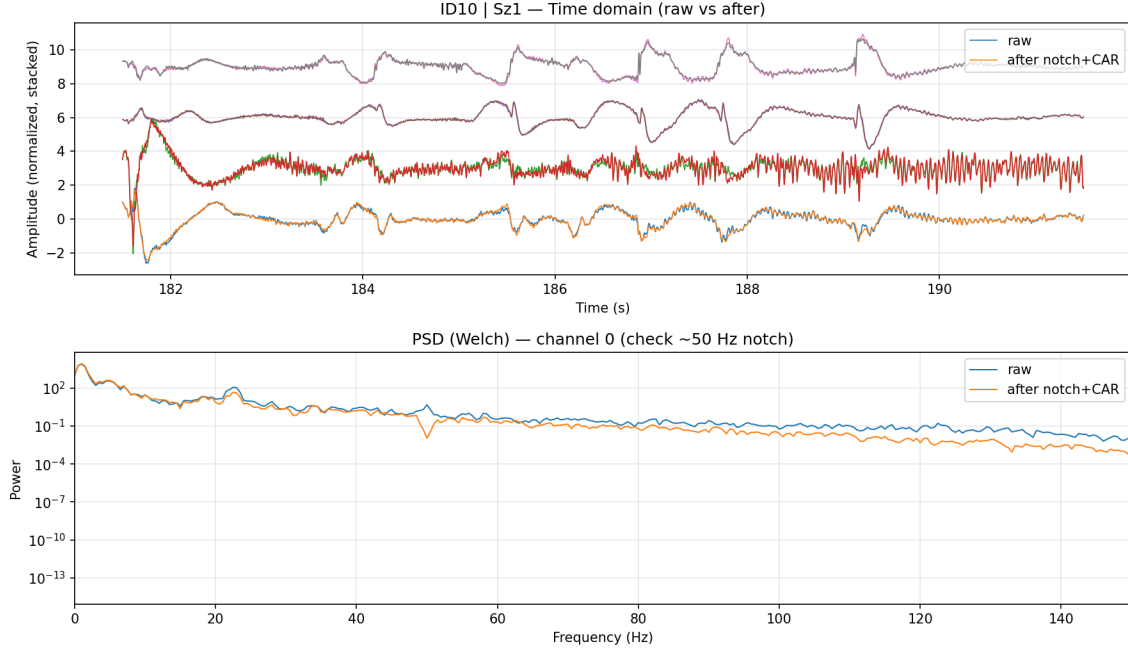


Figure 4-5: Results of pre-processing of SWEC ETHZ dataset

run-02.

The new pre-processing pipeline with subject HUP146 showed the following findings: run-01 has 50 epochs, 31 ripple, 199 FR, and 13 FRandR. Its top FR channels are led by LG8 and RI11, and the top FRandR channel is clearly LG8. The FR 95th-percentile area is LG8, RI11, and the FRandR 95th-percentile area is LG8.

Run-02 has 39 epochs, 21 ripple, 47 FR, and 5 FRandR. Its strongest FR channels are LG8 and RC5, and the top FRandR channels are RC5 and LG8. The FR 95th-percentile area is LG8, and the FRandR 95th-percentile area is RC5.

So, LG8 occurs in both runs at the FR and FRandR level, RC5 appears in both runs at the FR level, and becomes the top FRandR channel in run-02. Thus, the subject has stable candidates and cross-run recurrence of the same FR/FRandR channels.

4.2 Seizure detection

4.2.1 SWEC ETHZ dataset

Pre-processing results The result of SWEC ETHZ pre-processing can be seen

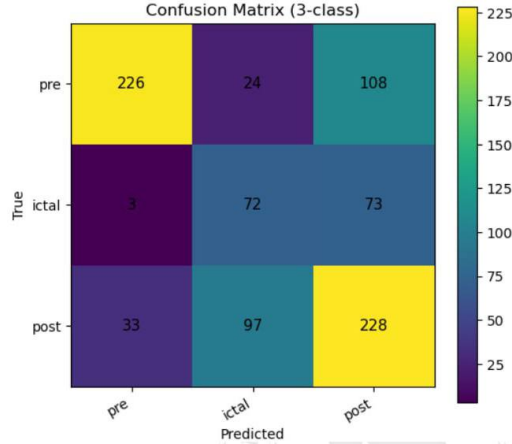


Figure 4-6: Confusion matrix of hybrid model with CNN and BiGRU

in Figure 4-5. The plot shows time series and PSD graphs before and after pre-processing. As it was mentioned before, the dataset was already pre-processed by the authors, so the effect is less visible compared to other pre-processing results. Nevertheless, it can be seen from the PSD plot that there is a clear peak at 50 Hz due to the effect of the Notch filter.

Seizure detection results Regarding the results of the seizure detection model,

Table 4.4: Seizure detection performance by class

Class	Sensitivity	Specificity	Precision	F1
Pre-ictal	0.63	0.93	0.86	0.73
Ictal	0.486	0.83	0.373	0.422
Post-ictal	0.64	0.64	0.557	0.594

Table 4.4 illustrates the results for three-class classification with sensitivity, specificity, precision, and F1 score for each class. The mean accuracy resulted in 0.58. The confusion matrix can be seen in Figure 4-6.

So, comparing the results in the Table 4.4, it can be concluded that the pre-ictal class scores are higher compared to the ictal and post-ictal classes' results. It can be explained by the labeling technique used in the dataset, where the first 3 s of recording is kept for the pre-ictal segment, 3 s till the end of recording is kept for the post-ictal segment, while the rest in between is the ictal segment that can vary in duration. So, the classifier performs well with the pre-ictal segment, while it

struggles to differentiate ictal and post-ictal segments. The ictal class results are also smaller than post-ictal class results; thus, mostly the ictal class samples are treated as post-ictal class samples by the classifier. The same conclusion can be made from the confusion matrix shown in Figure 4-6.

Therefore, the model was changed in the following way. Firstly, the windowing was changed to a 2-s sliding window and variable hop (preictal = 1 s, postictal = 1 s, ictal = 0.25 s) for training and fixed hop 1-s for testing, following the method used in Wang et. al. (2021) [26]. Secondly, the model was changed to a lightweight network with three layers of 2D CNNs. The kernel size was (3x3), and the padding was 1. Each CNN layer had a batch normalization layer, a ReLU activation function, and a max pooling layer. At the last 2D CNN network, there was an AdaptiveAvgPool layer instead of a MaxPool layer. The result then went to flattening layer and linear layers that output 3 logits (0 - pre-ictal, 1 - ictal, 2 - post-ictal).

The training procedure was almost the same as in the first attempt, except for the addition of early stopping at 7. Also, the training procedure included the leave-one-seizure-out (LOSO) method, meaning that one patient (ID1) is trained on 12 seizures and tested on 1 held-out seizure, which is repeated for 13 folds following the example from work of Wang et. al. (2021) [26].

In such a way, the mean accuracy was increased to 0.705, but the problems with ictal and post-ictal classes' misclassification still remained the same. Therefore, the choice of the dataset later shifted to the HUP dataset since the three-class labeling in the SWEC ETHZ dataset does not represent a real-world case.

4.2.2 HUP dataset

Pre-processing results The results are shown in Figure 4-7 with the pre-processing effect for run01 and Figure 4-8 with the pre-processing effect for single channel of run01. It can be noticed that the plots after pre-processing are less noisy in both figures. The Figure 4-8 demonstrates a smaller amplitude range of the signal, indicating the effect of CAR.

Seizure detection results (ver1) Now, the results for binary classification using

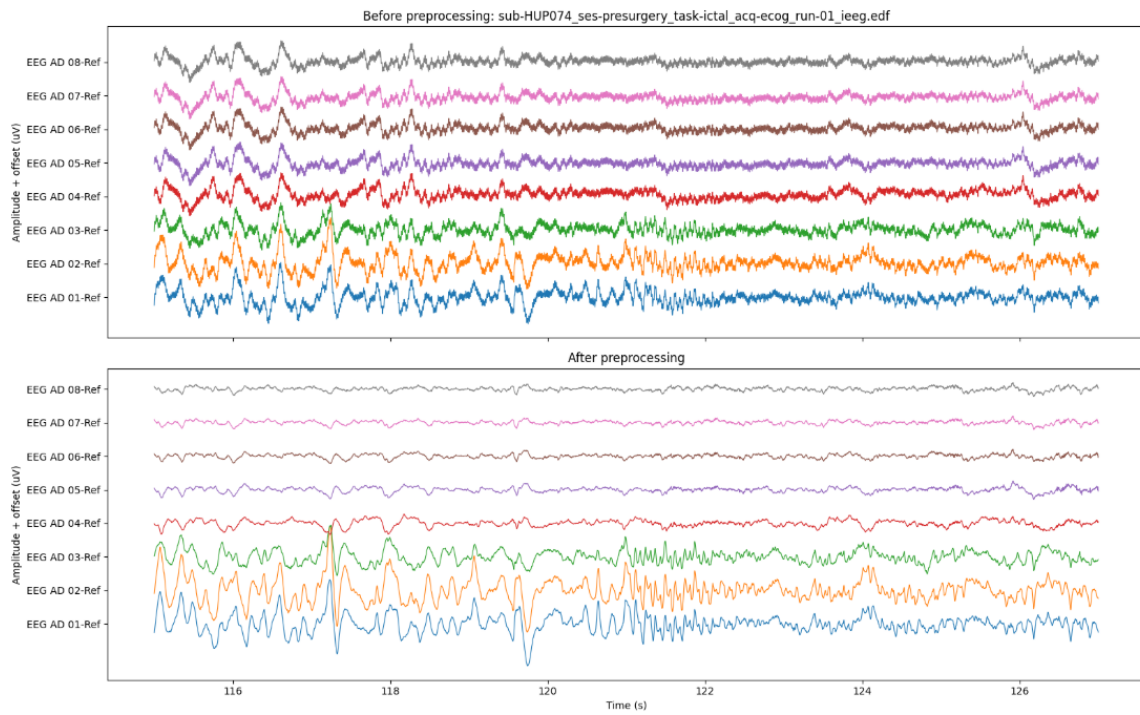


Figure 4-7: Before and after pre-processing for run01

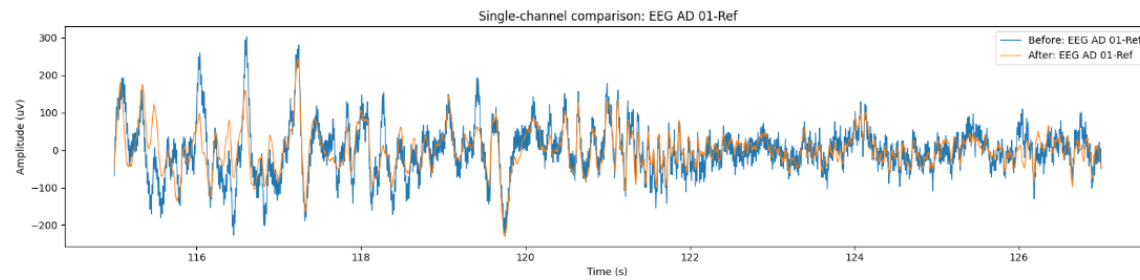


Figure 4-8: Before and after pre-processing for the single channel of run01

the 1D CNN network with channel attention network can be seen in Table 4.5.

Table 4.5: HUP dataset: seizure detection results (CNN+transformer)

Fold	Test Acc	Precision	Recall	F1	AU ROC
0	0.72	0.915	0.648	0.76	0.82
1	0.99	0.996	1.0	0.99	0.99
2	0.98	0.975	1.0	0.987	0.99
3	0.979	0.99	0.97	0.98	0.99
4	0.987	0.997	0.98	0.99	0.99

The most noticeable result is the difference in the test accuracy for Fold 1 compared to other folds in Table 4.5. The recall and F1 scores are also significantly lower for Fold 1 compared with other folds. The confusion matrix for Fold 1 has the following values: $TN = 155$, $FP = 23$, $FN = 133$, $TP = 245$. So, the number of FN is much larger compared to FP, meaning that the model is missing many seizure windows. At the same time, the model seems to overfit, looking into the rest of the results in Table 4.5.

Therefore, as the second attempt, the model was changed to a smaller one with CNN hidden channels reduced from 64 to 32, embedding dimensions reduced from 128 to 64, transformer heads reduced from 4 to 2, and transformer layers from 2 to 1 to deal with overfitting and reduce the number of FN. Also, the regularization was changed with dropout increasing from 0.2 to 0.3 and early stopping decreasing from 8 to 5. Threshold was also changed from a fixed value of 0.5 to a threshold tuning method, where values varied as in the following range for each fold: (0.2, 0.25, 0.3, ..., 0.5). For the threshold-tuned experiments, the decision threshold was selected only on the validation split within each fold by maximizing validation performance over a predefined threshold grid. The test fold was then evaluated once using the threshold selected from validation data. Therefore, threshold tuning was performed without access to test labels. The corresponding value of the threshold for each run was determined by the calculation of F1 scores. In such a way, the results with the new column showing the threshold for each fold are shown in Table 4.6.

The test accuracy for Fold 1 increased from 0.72 to 0.793, recall increased from 0.648 to 0.912, and F1 remained unchanged. Also, comparing the values from the

Table 4.6: HUP dataset: seizure detection results (smaller model)

Fold	Test acc	Precision	Recall	F1	AU ROC	Threshold
1	0.793	0.915	0.912	0.76	0.88	0.35
2	0.996	0.994	0.993	1.0	0.99	0.25
3	0.99	0.991	0.984	1.0	0.99	0.2
4	0.974	0.975	0.99	0.96	0.99	0.45
5	0.99	0.99	0.994	0.992	0.99	0.2

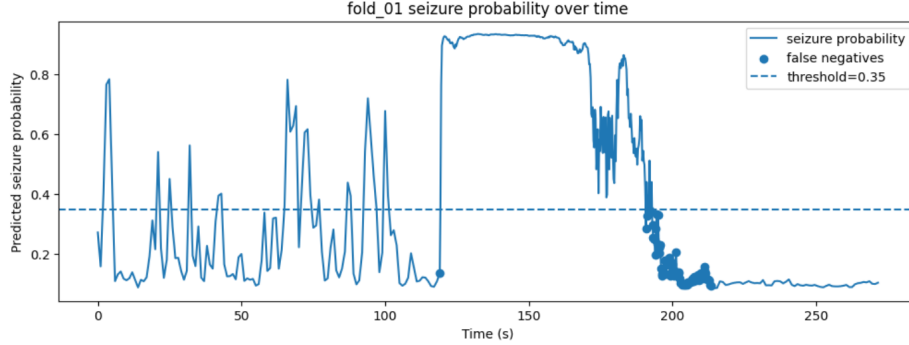


Figure 4-9: Seizure probability plotted over time (Fold 1)

confusion matrix ($TN = 150$, $FP = 28$, $FN = 87$, $TP = 291$), the number of FN decreased from 133 to 87. In addition, the rest of the results for other folds are now not close 100% performance, which is the result of stronger regularization.

Nevertheless, the fold 1 test accuracy remained much smaller compared to other fold results. Thus, to understand the problem, seizure probabilities were plotted over time for Fold 1 in Figure 4-9. It can be seen that the seizure is detected by the classifier in a time window from approximately 120 to 170 s. But after that, there are many FN points left under the threshold line. So, FN points are concentrated near seizure onset, late ictal, or near offset; thus, it can be hypothesized that longer windows may provide more temporal context for transition regions.

So, as the third attempt, the windowing was changed from 2 s to 3 s segments following the example of Sun et al. (2025), who used the same 3 s window with the HUP dataset [23]. The attempt finally led to good results with a test accuracy of 0.95 for Fold 1, which is shown in Table 4.7. In addition, given the confusion matrix for this attempt ($TN = 173$, $FP = 3$, $FN = 22$, $TP = 359$), the number of FN decreased from 87 to 22 compared to the previous attempt, which is a significant result.

Table 4.7: HUP dataset: seizure detection results (3 s window)

Fold	Test acc	Precision	Recall	F1	AU ROC	Threshold
1	0.95	0.99	0.93	0.96	0.98	0.3
2	0.996	0.994	1.0	0.99	0.99	0.35
3	0.98	0.97	1.0	0.98	0.98	0.3
4	0.96	0.964	0.99	0.96	0.97	0.2
5	0.98	0.99	0.979	0.987	0.99	0.45

To conclude on seizure detection, the HUP dataset was chosen over the SWECETHZ dataset because it has more subject data and more accurately reflects real-world scenarios. The training of the HUP dataset finished with Fold 1 having worse performance compared to the other folds. The regularization, threshold tuning, and changed windowing methods were used to increase the performance of the model.

The next point was to make a cross-subject analysis to see the performance of the model for other subjects. The results of the cross-subject analysis were saved to a .csv file and reported the number of runs, number of ictal, non-ictal, and total windows, number of unique samples and folds, mean test accuracy, mean test balanced accuracy, test recall, precision, F1, and AUROC for each subject. The results for some subjects are presented in Table 4.8. Several conclusions can be made: first, some subjects have balanced accuracy equal to 0.5 and generally low performance, which cautious problem pointing to a class imbalance; second, the fraction of ictal and non-ictal samples varies severely among subject some having more of ictal samples and other having more non-ictal samples; third, the number of unique samples also varies, leading to a slow training process among subjects with a large number of samples.

Table 4.8: Subject-wise seizure detection results on the HUP dataset.

Subject	Samples	Windows	Ictal	Non-ictal	Folds	Acc.	Bal. Acc.	Prec.	Rec.	F1	ROC
HUP065	1536	2188	1067	1121	1	0.974	0.960	0.963	1.000	0.981	0.989
HUP070	1536	1524	137	1387	5	0.453	0.662	0.362	0.947	0.449	0.752
HUP074	1536	2967	1492	1475	5	0.976	0.978	0.990	0.971	0.980	0.992
HUP075	1536	3427	2677	750	1	0.938	0.500	0.938	1.000	0.968	0.554
HUP080	1536	2372	1072	1300	1	0.943	0.935	0.944	0.966	0.955	0.984

Subject	Samples	Windows	Ictal	Non-ictal	Folds	Acc.	Bal. Acc.	Prec.	Rec.	F1	ROC
HUP086	1536	1496	614	882	1	0.649	0.500	0.649	1.000	0.787	0.936
HUP087	1536	1321	373	948	1	0.611	0.500	0.611	1.000	0.759	0.962
HUP088	1536	3639	2733	906	1	0.817	0.909	1.000	0.817	0.899	0.908
HUP089	1536	2529	1373	1156	1	0.743	0.601	0.727	0.997	0.841	0.882
HUP133	1536	3138	1677	1461	5	0.951	0.959	0.989	0.934	0.960	0.981
HUP134	3072	1097	325	772	1	0.649	0.500	0.649	1.000	0.787	0.213
HUP135	1536	2286	1549	737	1	0.994	0.500	0.994	1.000	0.997	0.000
HUP139	3072	1729	614	1115	1	0.711	0.500	0.711	1.000	0.831	0.437

Seizure detection results (ver2) In order to deal with the problems, some inspection needed to be made. The main reason for the different number of unique samples is sampling frequency, which is different for each subject. Thus, all subjects were resampled to 128 Hz to avoid a varying number of samples across subjects. Next, the problem of varying samples of ictal/non-ictal classes was tried to be solved by inspecting the source files, such as events.tsv channels.tsv, electrodes.tsv, and ieeg.edf. It was found that the parser did not identify ictal windows correctly, and for some subjects, the number of identified ictal windows could be equal to 0, although events.tsv had ictal windows recorded. Later, it was determined that the parser used to read only seizure duration from the source files, while missing onset/offset markers. Therefore, the parser was changed to read sz onset/offset pairs. Also, the pre-processing effect could be too harsh, because the number of ictal windows dropped after the pre-processing. So, artifact rejection `z_threshold` was changed from 5 to 7, while keeping non-ictal strict to soften the effect of pre-processing, and ambiguous windows near seizure boundaries were ignored during training. The windowing was set as follows (window equals 3 sec):

```
TRAIN_HOP_NONICTAL_SEC = 1.0
```

```
TRAIN_HOP_ICTAL_SEC = 0.5
```

```
TEST_HOP_SEC = 1.0
```

One of the best improvements was the reduced time of training for subjects with a

large number of samples. In addition, the Fold 1 now performs similarly well as other folds, which is also a good sign. This time, cross-subject was made per run, to see the performance of each run separately instead of the mean performance across runs for a specific subject. After inspecting the .csv file, the following conclusions were made: 91 out of 201 runs had test balanced accuracy or test accuracy below 80%, out of which 23 runs had either of the accuracies below 50%. Some runs were equally bad for the whole subject, while several subjects had both good and bad runs. Also, the subject HUP146 that was held out for LLM testing showed poor results, which are shown in Table 4.9.

Table 4.9: Run-level seizure-detection results for subject sub-HUP146 seizure-detection version 2 (ictal run-03)

Subject	Accuracy	Balanced Acc.	Precision	Recall	F1-score	AUROC	AUPRC
sub-HUP146	0.681	0.713	0.583	0.993	0.734	0.993	0.994

Seizure detection results (ver3)

To find the root of the problem events.tsv files and .npz files after the pre-processing (checked rms_z and peak_z separation) were analyzed. After inspecting events.tsv file, some runs were found to have a too-short ictal window. For example, run04 of subject HUP070 has the onset duration from 120 s to 124.998 s, with the number of ictal instances being 9 and the total number of windows being 187. However, most runs with low performance have long enough ictal windows, so there could be several problems leading to low performance. Few runs were taken for analysis:

- Bad runs with long seizure window: HUP111 run01, HUP181 run02/run03, HUP173 run01
- Bad runs with short seizure window: HUP074 run04/05
- Good runs with long seizure window: HUP074 run01, HUP146 run01, HUP138 run02

The run01 of subject HUP074 was compared with run02 of subject HUP181. The Figure 4-10 shows a plot of rms_z and peak_z over time for subject HUP074, and

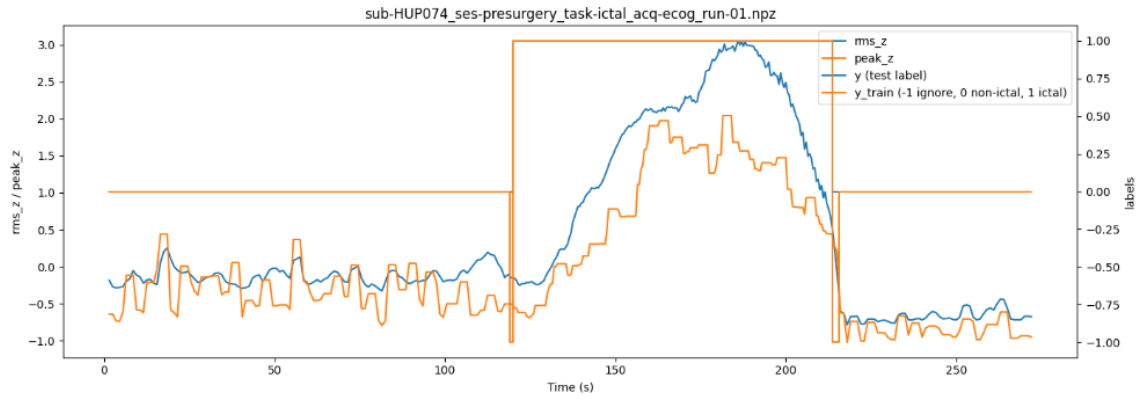


Figure 4-10: HUP074 run01 rms_z and peak_z plotted over time

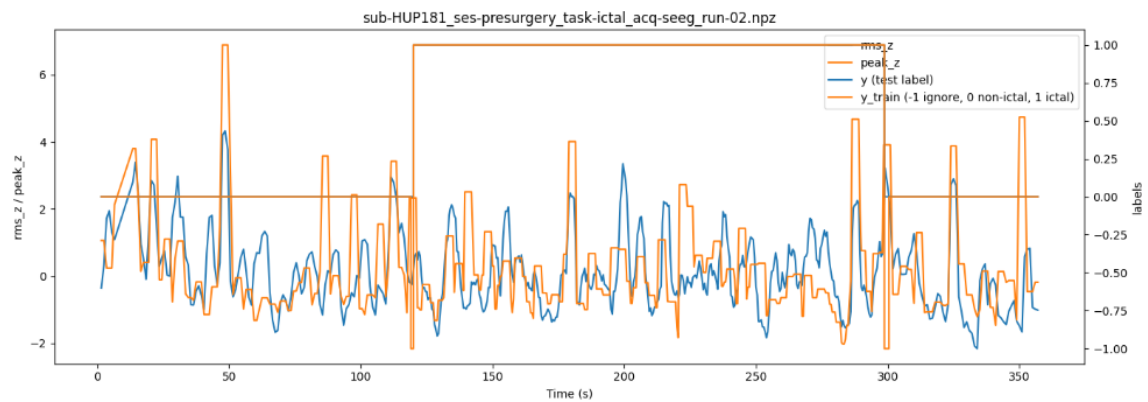


Figure 4-11: HUP181 run01 rms_z and peak_z plotted over time

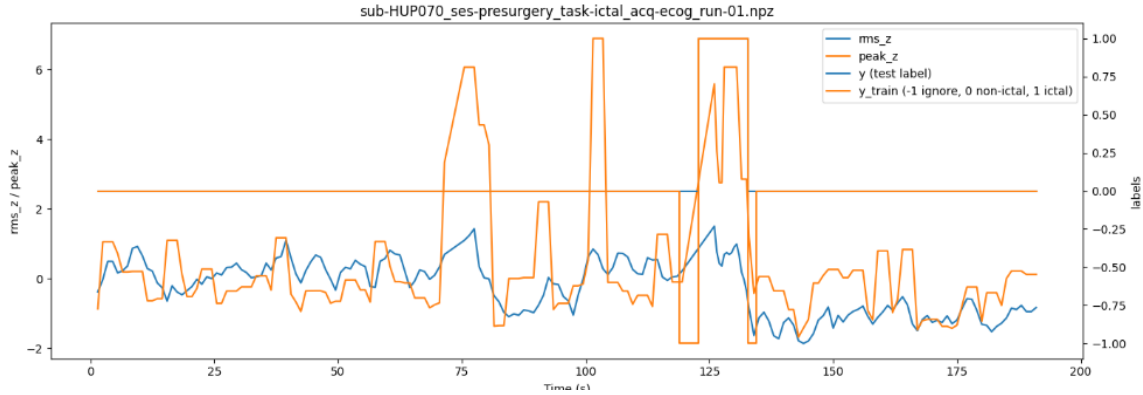


Figure 4-12: HUP070 run01 rms_z and peak_z plotted over time

Figure 4-11 shows a similar plot for subject HUP181. Comparing the plot side by side, it is obvious that before seizure, rms_z and peak_z mostly stay near 0, and both increase during seizure for subject HUP074. Also, this subject shows good class separability:

rms_z: non-ictal -0.29 vs ictal 1.66

peak_z: non-ictal -0.47 vs ictal 0.89

Meanwhile, Figure 4-11 shows an abnormal pattern with large spikes appearing before, during, and after seizure. Additionally, non-ictal windows are sometimes even larger than ictal windows, meaning the class boundary is broken after preprocessing. The class separability for this subject is very poor, which also supports the previous findings:

rms_z: non-ictal 0.22 vs ictal 0.11

peak_z: non-ictal 0.48 vs ictal 0.22

Additionally, a similar comparison was made for one subject case, Figures 4-12 and 4-13 show run01 and run04 for subject HUP070. In this case, both runs have good subject separability, but run01 has 14 ictal windows and run04 has 9 ictal windows. Therefore, for this case, the problem is in short ictal windows. The rms_z and peak_z values for run01 of subject HUP070 were as follows:

rms_z: -0.30 vs 0.59

peak_z: 0.18 vs 4.59

While the run04 of subject HUP070 had the following values for rms_z and peak_z:

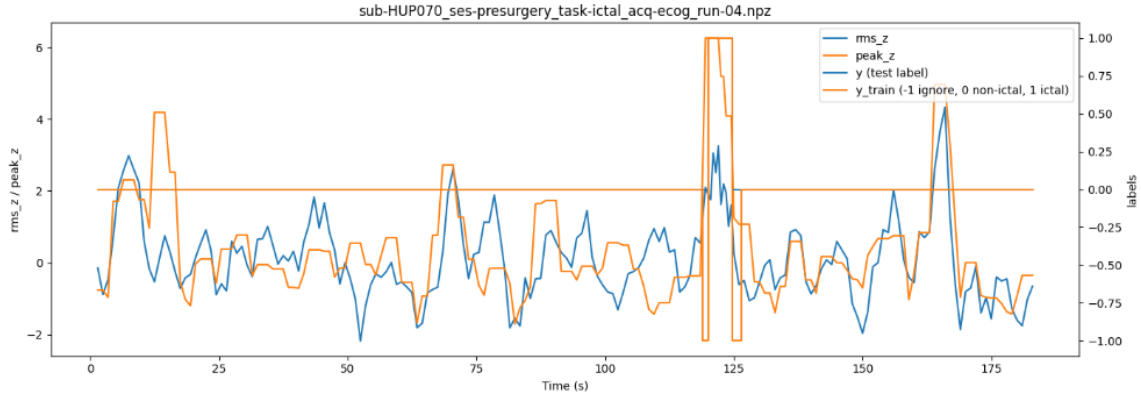


Figure 4-13: HUP070 run04 rms_z and peak_z plotted over time

rms_z: 0.06 vs 2.11

peak_z: 0.28 vs 5.30

So, the following changes were made for the final version of the seizure detection model:

- Some runs with too short ictal windows were skipped
- Threshold grid was changed from 0.1 to 0.91 with a 0.05 step
- Subjects had problems with channel/reference/modality variety across subjects, i.e., some subjects may use ECoG and others SEEG. It was fixed by introducing a 128-channel representation with masked padding
- The class separation problem was used by per-subject sample retention, i.e., 200 samples per subject, similar to the work of Sun et. al. (2025) [23], since some subjects also had a reversed class distribution problem (number of ictal windows was larger than non-ictal windows)
- Band-pass filter was changed from 0.5-120 Hz to 0.5-60 Hz
- Interictal runs were removed from the training stage, because previous results showed a high count of FP

After the changes, the final seizure model performs well for most of the subjects. Across 43 subjects, the mean test accuracy and mean balanced accuracy were 0.832

and 0.833, respectively. The mean precision, recall, and F1-score were 0.790, 0.830, and 0.781, while the mean AUROC and AUPRC reached 0.900 and 0.885, respectively. The corresponding median values were consistently higher, with a median accuracy of 0.872, median balanced accuracy of 0.873, median F1-score of 0.840, median AUROC of 0.970, and median AUPRC of 0.955. This difference between the mean and median indicates that the model performed well for many subjects, whereas a smaller group of difficult cases with low scores reduced the overall average.

Table 4.10 represents examples of strong, moderate, and difficult or unstable subjects. Among the strongest cases, subjects sub-HUP160, sub-HUP088, and sub-HUP142 achieved near-perfect performance, with F1-scores of 0.995, 0.992, and 0.988, respectively. These results demonstrate that the final model can detect ictal activity very reliably for a subset of subjects. Moderate subjects, such as sub-HUP074, sub-HUP080, and sub-HUP106, also showed consistently high performance, with F1-scores between 0.956 and 0.972, confirming that the model generalized well beyond only the best cases. At the same time, the results reveal several difficult and unstable subjects. In particular, sub-HUP117 and sub-HUP181 showed weak performance, with F1-scores of 0.277 and 0.418, respectively. Subject sub-HUP070 achieved higher average accuracy (0.809) and balanced accuracy (0.838), but its mean F1-score remained substantially lower (0.601), and its standard deviation of F1-score was high (0.310), indicating strong variability across runs. This pattern supports the conclusion that seizure detection on the HUP dataset is strongly influenced by subject-to-subject and run-to-run variability, rather than showing uniformly stable performance across the cohort.

The distribution of results across all 43 subjects further supports this interpretation. Out of 43 subjects, 29 achieved a mean accuracy of at least 0.80, 31 achieved a mean balanced accuracy of at least 0.80, and 26 achieved mean F1-score of at least 0.80. In addition, 16 subjects reached a mean accuracy of at least 0.90, and 10 subjects exceeded 0.95 accuracy. Talking run-wise so that it can be compared to seizure detection version 2 results, 67 out of 182 runs had test-balanced accuracy or test accuracy below 80%, 19 of which had either of the accuracies below 50%. Many runs

Table 4.10: Representative subject-level results of the final seizure-detection model on the HUP dataset. Subjects are grouped into strong, moderate, and difficult/unstable cases.

Group	Subject	Runs	Accuracy	Balanced Acc.	F1-score	AUROC	Std F1
Strong	sub-HUP160	3	0.997	0.996	0.995	1.000	0.005
Strong	sub-HUP088	3	0.989	0.988	0.992	0.999	0.005
Strong	sub-HUP142	3	0.989	0.988	0.988	0.999	0.004
Moderate	sub-HUP074	5	0.972	0.974	0.968	0.997	0.023
Moderate	sub-HUP080	4	0.976	0.978	0.972	0.999	0.019
Moderate	sub-HUP106	3	0.956	0.958	0.956	0.980	0.019
Difficult / unstable	sub-HUP070	5	0.809	0.838	0.601	0.904	0.310
Difficult / unstable	sub-HUP117	3	0.652	0.543	0.277	0.450	0.440
Difficult / unstable	sub-HUP181	5	0.431	0.510	0.418	0.428	0.282

(24 out of 47) had an AUROC score of about 0.9, but balanced accuracy is a bit below 0.8.

The remaining problem is a few subjects, such as HUP173, HUP181, HUP111, etc., that have true low class separability. Another important observation is the presence of unstable subjects, for which performance varied substantially across runs. The highest standard deviations of F1-score were observed for sub-HUP123 (0.446), sub-HUP117 (0.440), sub-HUP162 (0.387), sub-HUP070 (0.310), and sub-HUP181 (0.282). This indicates that for some patients, the model is not only weaker on average, but also less consistent from run to run. Therefore, the fairest interpretation is that the final model achieved good to excellent performance for a substantial subset of subjects, while still showing weak generalization in several challenging cases.

For the downstream LLM pipeline, subject sub-HUP146 was used as the main case-study subject. As shown in Table 4.11, the reserved-run seizure-detection result for this subject was strong. In comparison with the previous version’s results for the held run in Table 4.9, the final version achieved substantially better performance for the HUP146 case-study subject, improving accuracy from 0.681 to 0.963, balanced accuracy from 0.713 to 0.928, precision from 0.583 to 1.000, and F1-score from 0.734 to 0.922, while maintaining very high AUROC and AUPRC. These results support the use of HUP146 as a suitable showcase subject for the LLM-based reporting pipeline. However, this result should be seen as a case-study example rather than as representative of all subjects.

Table 4.11: Run-level seizure-detection results for subject sub-HUP146 seizure-detection final version

Subject	Accuracy	Balanced Acc.	Precision	Recall	F1-score	AUROC	AUPRC
sub-HUP146	0.963	0.928	1.000	0.855	0.922	1.000	1.000

4.3 LLM

4.3.1 Version 1

The version 1 unified LLM system was tested on subject sub-HUP146 using one merged HFO subject-level JSON together with two reserved seizure JSONs, corresponding to one ictal and one interictal run. The expectation from the system was not to assess raw event detection directly, but to evaluate whether the LLM could correctly integrate outputs from both branches, keep important information, avoid unsupported conclusions, and generate a coherent report. The raw event detection is handled by deterministic signal-processing and machine-learning pipelines, i.e., HFO detection and seizure detection branches. Importantly, the provided results for Version 1 use the seizure detection version 1 results. The rest of the versions of the LLM system development have updated results with version 3 of seizure detection.

Table 4.12: Summary of structured HFO output exported for subject sub-HUP146.

JSON section	Stored information
Subject metadata	Subject ID, dataset name, analyzed interictal runs
Ripple detections	Channel-wise ripple counts and rates per minute
Fast ripple detections	Channel-wise FR counts and rates per minute
FRandR pairs	Co-occurring ripple and fast ripple events per channel
HFO-area candidates	Channels above the 95th percentile of FR or FRandR rate
Quality-control fields	Epoch counts, rejected segments, and detection statistics

Table 4.12 shows the content of JSON for HFO detection on the example of sub-HUP146. The HFO detection distinguished between strong-consensus channels and moderate candidate channels. The strongest evidence was observed for channels LG8 and RC5, and additional moderate candidate channels included RI11, RJ5, LG7, LG5, and LG4. The aggregated HFO summary reported 89 usable epochs, 52 ripple events, 246 fast-ripple events, and 18 FRandR pairs, with no rejected burst rows.

Overall, the HFO branch produced a stable and clear subject-level summary.

Table 4.13 shows a content summary of two JSON files for seizure detection for ictal and interictal runs. The seizure detection was evaluated on the held-out runs from training. For the ictal reserved run (sub-HUP146_ses-presurgery_task-ictal_acq-seeg_run-03), the final exported JSON contained 347 windows, of which 166 were labeled ictal and 181 nonictal. Using a decision threshold of 0.8, the model predicted 142 positive windows, corresponding to a predicted-positive fraction of 40.9%, with a mean smoothed probability of 0.543 and peak smoothed probability of 0.909. The predicted ictal activity was concentrated in two candidate segments spanning approximately 119.0-132.5 s and 136.5–199.0 s, suggesting that the model did not mark the entire run diffusely as ictal, but instead focused on a restricted part of the recording. For the interictal reserved run (sub-HUP146_ses-presurgery_task-interictal_acq-seeg_run-02), the final JSON contained 298 windows, all labeled non-ictal. At the same threshold of 0.8, the model predicted 0 positive windows, with a mean smoothed probability of 0.251 and a peak smoothed probability of 0.306. Consequently, no candidate ictal segments were produced for the interictal run.

Table 4.14 demonstrates a summary of the reasoning JSON file. Reading through the reasoning JSON shows that LLM correctly preserved the distinction between strong-consensus HFO channels (LG8, RC5) and moderate candidate HFO channels (RI11, RJ5, LG7, LG5, LG4). Also, LLM correctly recognized that patient-level fusion was supported, seizure channel-level evidence was unavailable, direct HFO–seizure channel concordance was therefore not supported, and seizure detection in the final configuration had a low false-positive burden. In general, the interpretation demonstrates desirable behavior because LLM followed the limitations written in the integration layer rather than over-interpreting the data. The system could compare HFO findings with seizure-run behavior at the subject level, but it did not claim direct spatial overlap between HFO channels and seizure onset, since the seizure branch did not provide channel-level localization.

Figure 4-14 shows the final report for version 1, which is technically coherent and follows the specified structure, including sections such as system summary, HFO

Table 4.13: Summary of the seizure-detection JSON output exported for subject sub-HUP146.

JSON section	Stored information
Subject and run meta-data	Subject identifier, dataset name, task type, acquisition type, and analyzed ictal or interictal run.
Model and inference settings	Trained seizure-detection model used for inference, decision threshold, window length, hop size, and probability-smoothing settings.
Window-level predictions	Number of analyzed windows, number of true ictal and non-ictal windows when labels were available, predicted-positive windows, and predicted-positive fraction.
Probability statistics	Mean and peak seizure probability across the analyzed run, usually computed after temporal smoothing.
Candidate ictal segments	Start time, end time, duration, number of positive windows, mean probability, and peak probability for detected candidate seizure segments.
Per-window output	Time-resolved seizure probabilities and binary predictions for each window, allowing visualization of seizure probability over time.
Channel-level evidence	Model-derived channel contribution scores for early ictal windows, reported as gradient-input saliency proxies rather than direct electrophysiological onset annotations.
Channel-name mapping	Mapping between internal model channel indices and clinically interpretable bipolar electrode names, allowing comparison with HFO candidate channels.
Role in LLM pipeline	Provides structured seizure-detection evidence used by the LLM system for report generation and cross-modal comparison with HFO findings.

Table 4.14: Summary of the Version 1 reasoning JSON generated for subject sub-HUP146.

JSON section	Purpose in the reasoning output
Case assessment	Provides an overall description of the analyzed subject case and identifies the available structured inputs from the HFO and seizure-detection branches.
HFO assessment	Summarizes HFO evidence, including candidate HFO-area channels, FR or FRandR findings, and channel-level biomarker information.
Seizure assessment	Summarizes seizure-detection results, including detected candidate ictal segments, seizure-probability information, and run-level model findings.
Integration assessment	Provides an initial comparison between HFO findings and seizure-detection findings. In Version 1, this comparison was mainly LLM-generated and did not yet rely on deterministic cross-modal fusion.
Limitations	Records uncertainty related to model predictions, absence of direct clinical interpretation, possible mismatch between modalities, and the need to avoid overconfident localization.
Final interpretation	Produces a concise text-based conclusion based on the structured HFO and seizure evidence. This interpretation was later used to generate the final report.
Role in the pipeline	Acts as an intermediate reasoning layer between the compact case JSON and the final generated report.

module summary, seizure module summary, integration constraints, overall interpretation, and limitations. Overall, the results of version 1 prove that LLM is capable of performing unified multimodal reasoning. The LLM did not hallucinate seizure onset channels, did not claim unsupported HFO–seizure spatial concordance, and produced an uncertainty-aware report that corresponds with the actual structured inputs. However, one of the major problems is the absence of channel-level seizure onset and spread information, and LLM could not perform true channel-level multimodal concordance analysis. Also, the final report highlights that LLM mainly copied input JSON without much reasoning. Thus, the next version was mainly concentrated on these problems.

Version 1 final report for subject HUP146

System Summary

- Supports patient-level fusion: **True**
- HFO channel-level evidence available: **True**
- Seizure channel-level evidence available: **False**
- Direct HFO-seizure channel concordance supported: **False**
- Recommended interpretation mode: **Limited**
- System confidence: **Limited** due to absence of seizure channel-level evidence and inability to establish direct HFO-seizure concordance.

HFO Module Summary

- **Strong consensus channels:** LG8, RC5, high agreement across analyses.
- **Moderate consensus channels:** RI11, RJ5, LG7, LG5, LG4, lower agreement but still notable.
- **Key findings:** LG8 and RC5 show robust HFO activity; RI11, RJ5, LG7, LG5, and LG4 show moderate activity with variable consistency.
- **Limitation:** Direct HFO-seizure channel concordance **cannot be established** due to lack of seizure channel-level evidence.

Seizure Module Summary

- **Ictal run:** 142 true ictal windows and 178 true nonictal windows; 75.6% predicted positive.
- **Interictal run:** High false-positive burden, 297 predicted positive windows out of 298 total.
- **Key findings:** Significant interictal false positives reduce confidence in seizure-channel associations; ictal activity is localized to 2.0–262.5 seconds.
- **Limitation:** No channel-level seizure onset or spread evidence is provided.

Integration Constraints

- Seizure JSON lacks channel-level onset or spread evidence.
- Direct HFO-seizure channel concordance **cannot be established** without seizure channel-level data.
- HFO findings can be compared with seizure run behavior at the subject level, but **no spatial overlap claims** can be made.

Overall Interpretation

- Strong HFO consensus on **LG8** and **RC5**; moderate consensus on **RI11**, **RJ5**, **LG7**, **LG5**, and **LG4**.
- Seizure detection has a **high interictal false-positive burden**, limiting confidence in direct HFO-seizure channel associations.
- No channel-level seizure onset or spread evidence is available to support spatial overlap claims.

Limitations

1. Seizure channel-level evidence is **unavailable**.
2. Direct HFO-seizure channel concordance **cannot be established** due to missing data.
3. High interictal false-positive burden in seizure detection further reduces confidence in associations.

Figure 4-14: Excerpt from the Version 1 generated report for subject HUP146.

4.3.2 Version 2

This version improved data representation by introducing real bipolar seizure channels, channel-level seizure support evidence, and deterministic integration outputs, as it was described in the Methods section. For subject HUP146, the final compact representation showed: strong HFO consensus in channels LG8, RC5, RI11, RJ5, LG7, LG5, LG4, ictal seizure segments at 119.0–132.5 s and 136.5–199.0 s, seizure saliency-proxy channels centered around RG1-RG2, RG3-RG4, RH1-RH2, RG2-RG3, RG5-RG6, RD3-RD4, RA2-RA3, and RA1-RA2, no direct cross-modal overlap (concordance_level = "none"), and a cross-modal mismatch warning with moderate integration confidence and cautious overall consistency. Therefore, the deterministic integration layer shows expected results. However, the problem with LLM appeared at the stage of transforming structured inputs into reasoning outputs. In order to deal with the problem, system prompts were changed several times alongside testing

with several LLM models.

Version 2 final report for subject HUP146 using Qwen3:8b

Analysis Summary

The provided JSON outlines HFO candidate channels and seizure-related saliency proxy channels, with no direct overlap between the two sets. Key findings include:

1. HFO Candidate Channels

- Identified as LG8, RC5, RI11, RJ5, LG7, LG5, LG4 with metrics like mean probability, `mean_prob_smooth`, and peak probability, `peak_prob_smooth`.
- These channels show moderate to strong HFO evidence but lack direct alignment with seizure-related saliency proxy channels.

2. Seizure Saliency Proxy Channels

- Highlighted as RG1-RG2, RG3-RG4, RH1-RH2, etc., with early window hits and mean saliency scores.
- These channels are tied to seizure onset via gradient-based saliency proxies, not direct electrophysiological annotations.

3. Deterministic Integration Results

- **No overlap:** the `hfo_candidate_channels` and `seizure_candidate_channels` share no common channels, with `overlap_count`: 0.
- **Concordance level:** “None,” indicating no direct electrophysiological overlap.
- **Consistency flags:** a “`cross_modal_mismatch`” warning is raised, emphasizing the absence of direct HFO-seizure channel alignment.

4. Integration Constraints

- Seizure evidence is based on supportive model-derived saliency proxies, not definitive localization.
- Disagreements between modalities, HFO versus seizure onset, should not be forced into a single conclusion but instead noted as cautious.

Conclusion

The analysis identifies supportive model-derived evidence for HFOs and seizure saliency proxies but highlights no direct electrophysiological overlap. The lack of concordance suggests the need for further investigation into indirect associations or alternative explanations, adhering strictly to the observed data without speculative conclusions.

Figure 4-15: Excerpt from the Version 2 generated report for subject HUP146 with Qwen3:8B.

First one was Qwen3:8B, which produced the most practically usable final report

among tested models in terms of readability, which can be seen in Figure 4-15. Its output clearly described ictal and interictal run metrics, top candidate seizure segments, top early ictal channels, top channel onset scores, HFO candidate channels, absence of overlap between modalities, cross-modal mismatch, and cautious interpretation. It also preserved the caution that seizure evidence was based on a saliency proxy rather than direct electrophysiological onset. However, Qwen3:8B was weaker at strictly structured reasoning. When forced to produce a separate reasoning JSON, it often generated meta-commentary such as “Okay, let’s see...” instead of the requested schema. So, the model functioned more reliably in the role of summarizer and report writer rather than a strict reasoning engine.

Table 4.15: Summary of the Version 2 reasoning JSON generated for subject sub-HUP146 using Qwen3-30B-A3B-Instruct-2507.

JSON section	Stored information and interpretation role
Case overview	Summarizes the subject, available input files, and the purpose of the reasoning stage.
HFO evidence	Lists candidate HFO channels and summarizes channel-level FR, ripple, and FRandR findings used for epileptogenic-zone hypothesis formation.
Seizure evidence	Summarizes candidate ictal segments, seizure-probability behavior, and early ictal channel evidence from the seizure-detection branch.
Saliency-proxy information	Includes model-derived channel evidence based on gradient-input saliency. The output is explicitly treated as supportive model evidence, not as a true clinical seizure-onset annotation.
Deterministic channel concordance	Reports whether HFO candidate channels overlap with seizure-supportive channels after channel-name mapping. This section supports cross-modal comparison.
Consistency checks	Flags agreement, disagreement, or missing correspondence between the HFO and seizure branches. These flags are used to prevent overconfident interpretation.
Integration constraints	Defines restrictions for the LLM, including avoiding invented channels, avoiding unsupported seizure-onset claims, and lowering confidence when modalities disagree.
Reasoning output	Produces a structured interpretation based on the available evidence, although Version 2 still showed variability in reasoning quality and report structure.

Q5: What is the concordance level between HFO candidate channels and seizure saliency proxy channels?

A: The concordance level between HFO candidate channels and seizure saliency proxy channels is “none”.

This is explicitly stated in the `deterministic_integration.channel_concordance` section of the JSON, where:

- `overlap_channels` is an empty array, `[]`;
- `overlap_count` is 0;
- `concordance_level` is “none”.

Despite strong evidence in both modalities—HFOs detected in channels such as LG8, RC5, RI11, RJ5, LG7, LG5, LG4, and seizure onset inferred via saliency proxy in channels like RG1-RG2, RG3-RG4, RH1-RH2, RG2-RG3, and RG5-RG6—there is no direct spatial overlap between the two sets of channels. This lack of overlap triggers a cross-modal mismatch warning and contributes to the cautious overall consistency and moderate integration confidence.

Figure 4-16: Excerpt from the Version 2 generated report for subject HUP146 with Qwen3-30B-A3B-Instruct-2507.

The second one was the Qwen3-30B-A3B-Instruct-2507 model, which made it easier to build a reasoning JSON with the required sections after introducing validation and repair logic. The reasoning JSON had all of the needed sections, which are shown in Table 4.15. Nevertheless, the content quality remained poor, while the final report shown in Figure 4-16 performs much worse than the final report with the 8B report by not following the needed structure and being too generic. Therefore, the 30B model improved format compliance, but not necessarily reasoning quality.

Third, the Qwen3.5:35B model was chosen as a stronger reasoning model. The main reason for its usage was the larger context window, which was expected to improve both reasoning and the ability to preserve the required structure in the final report. However, the results were not better than those of the smaller models, and several problems were observed, including longer reasoning, increased latency, frequent timeouts, and formatting drift or visible artifacts, i.e., extra tokens appearing before the JSON output. Examples of these issues included repaired outputs with structures such as `{"reasoning": ..., "final_answer": ...}` instead of the desired schema,

as well as raw outputs with visible artifacts before the final JSON block. The timeout problem was addressed by increasing the timeout value, but this did not change the overall behavior of the case:

```
resp = requests.post(OLLAMA_URL, json=payload, timeout=1800)
```

Although the 35B model could be forced into acceptable behavior with repair logic, it did not produce a clear clinical-style structure compared with the simpler 8B report. The main tradeoff was that the larger model increased engineering complexity and runtime without producing proportionally better final reports. Thus, the Qwen3:8B was chosen as the main model for the LLM system.

4.3.3 Version 3

The main result of version 3 was the stabilization of the multimodal reasoning stage after the introduction of deterministic cross-modal integration in version 2. In the previous version, the system already had access to richer multimodal evidence, including HFO candidate channels, seizure saliency-proxy channels, deterministic concordance measures, and consistency checks. However, the final LLM-generated report still showed shallow reasoning or was insufficiently aligned with the evidence present in the compact case JSON. Therefore, version 3 has primarily focused on the improvement of the reliability of reasoning.

The first important result of version 3 was the successful introduction of a strict reasoning JSON schema together with validation and repair, which is mentioned in Table 4.16. This substantially improved the consistency during the reasoning stage. The deterministic integration layer successfully captured the absence of exact cross-modal concordance. In the validated reasoning JSON for sub-HUP146, the integration assessment reported `{"concordance_mode": "exact_channel_match", "concordance_level": "none"}`, and an empty `{"overlap_channels_list": []}`. Thus, although both modalities produced meaningful findings, the system correctly recognized that the HFO candidate channels and seizure supportive channels did not overlap directly under

exact matching. As a result, the output became easier to validate, compare across cases, and reuse for report generation.

Table 4.16: Summary of the Version 3 validated reasoning JSON generated for subject sub-HUP146.

Validated JSON section	Function in the final reasoning pipeline
Schema-valid case summary	Stores the subject-level context and confirms that the reasoning output follows the predefined JSON schema.
HFO findings	Provides structured HFO evidence, including candidate HFO-area channels and the biomarker basis for these candidates.
Seizure findings	Provides structured seizure-detection evidence, including candidate ictal segments, probability-based findings, and channel-level saliency-proxy evidence.
Cross-modal integration	Reports deterministic comparison between HFO candidate channels and seizure-supportive channels. For sub-HUP146, the system identified no exact channel-level overlap between the two modalities.
Concordance fields	Stores fields such as concordance mode, concordance level, and overlap channels, making the integration result explicit and machine-checkable.
Confidence rationale	Explains how the system confidence was affected by supporting evidence, mismatch between modalities, proxy-based seizure evidence, and other limitations.
Limitations	Lists constraints such as limited clinical validation, use of model-derived saliency rather than direct onset annotation, and incomplete agreement between modalities.
Final impression	Provides a cautious, evidence-grounded interpretation that can be rendered into a standardized report without requiring the LLM to freely rewrite the whole report.
Validation and repair role	Ensures that the reasoning output satisfies the expected schema before being used for deterministic report generation.

The second result of version 3 was improved evidence-grounding in the generated reasoning. The updated constraints required the model to explicitly mention concrete evidence factors when providing confidence rationale, include timing information when ictal candidate segments were present, provide limitations under mismatch or proxy-evidence conditions, and refer to the effect of cross-modal disagreement on confidence. Consequently, the generated reasoning provided the available multimodal evidence, rather than relying on generic summary language. This made the reasoning stage more interpretable.

The third major result of version 3 was the standardization of the final reporting stage,

which can be seen in Figure 4-17. In version 2, final LLM-generated reports could vary in structure and phrasing, which made output quality harder to control. In version 3, the final report was rendered from the validated reasoning JSON and compact case JSON into stable predefined sections such as case overview, HFO findings, seizure findings, cross-modal integration, limitations and confidence, and final impression. This produced more predictable outputs and improved the reproducibility of the final reporting stage.

Version 3 final report for subject HUP146

Case overview

Subject sub-HUP146 from the HUP dataset was analyzed with integrated HFO and seizure branches. Cross-modal comparison used exact channel-name matching.

HFO findings

HFO consensus strength was strong. Strong consensus channels were LG8, RC5. Additional supportive channels were RI11, RJ5, LG7, LG5, LG4. Strong HFO consensus in LG8 and RC5; 52 ripples and 246 fast ripples detected across 89 epochs.

Seizure findings

Two ictal candidate segments detected: 119.0–132.5 sec, 13.5 sec duration, and 136.5–199.0 sec, 62.5 sec duration, both with high mean and peak smoothed probabilities. Low interictal false-positive burden. Seizure channel-level evidence was treated as model-derived saliency proxy. Top supportive seizure channels were RG1-RG2, RG2-RG3, RD3-RD4, RA2-RA3, RA1-RA2. Seizure evidence is model-derived saliency proxy; top supportive channels include RG1-RG2, RG2-RG3, RD3-RD4, RA2-RA3, RA1-RA2.

Cross-modal integration

Concordance mode was exact_channel_match. Concordance level was none. Overlap channels: none. No direct channel overlap between HFO consensus channels and seizure supportive channels; cross-modal mismatch warning lowers confidence.

Limitations and confidence

Overall confidence was low. HFO consensus is strong in LG8 and RC5, but seizure evidence is a model-derived saliency proxy with no overlap with HFO channels; the cross-modal mismatch warning further limits confidence. No overlap between HFO consensus channels and seizure supportive channels. Seizure evidence is the model-derived saliency proxy, not direct electrophysiological detection.

Final impression

No concordance between HFO consensus and seizure evidence; hypothesis deferred pending fur-

ther integration.

Figure 4-17: Excerpt from the Version 3 generated report for subject HUP146.

Overall, the main result of version 3 was that the unified system became sufficiently stable to produce validated multimodal reasoning outputs and standardized reports. This version established a reliable baseline for later development.

4.3.4 Version 4

The main result of version 4 was the successful introduction of bounded tool orchestration in addition to the reasoning pipeline established in version 3. For sub-HUP146, the planner LLM read the compact case and generated a valid orchestration decision. The planner justified this by stating that the integration evidence contained consistency flags and `concordance_level` was none, and therefore, the integrated case, reasoning, and report should be refreshed. This demonstrates that the LLM was able not only to interpret the current case state, but also to decide what sequence of internal actions was required next.

In the successful orchestration run, the system executed the following sequence of tools shown in Table 4.17. The execution log printed during the model running also followed this sequence from Table 4.17. In such a way, the updated reasoning output passed validation. This shows that version 4 successfully extended the system from a fixed reasoning workflow into a workflow in which the planner LLM selects which refresh steps should be executed. The result was not only a refreshed final report, but also a full orchestration trace consisting of the validated plan, execution log, and orchestration state.

Another important result of version 4 was that the planner’s decision remained grounded in the multimodal evidence. The compact case used by the planner still described strong HFO consensus in LG8 and RC5, two ictal candidate segments, low interictal false-positive burden, and no exact channel overlap, together with a cross-modal mismatch warning. The planner correctly recognized this as a case requiring refresh and reran the reasoning and reporting stages. In this way, version 4 demon-

Table 4.17: Tool execution sequence in the Version 4 orchestration pipeline.

Step	Tool	Purpose	Output
1	Rebuild compact case	Prepare reduced LLM input	Compact case JSON
2	Run reasoning	Generate structured interpretation	Reasoning JSON
3	Validate reasoning	Check schema and constraints	Validated reasoning file
4	Render report	Produce final report text	Structured report

strated the intended role of the LLM as a controller of bounded analysis functions, rather than simply as a passive report generator.

The final report for version 4, shown in Figure 4-18, seems to be more explicit compared to previous versions. For example, it stated that HFO consensus was strong in LG8 and RC5, that 52 ripples and 246 fast ripples had been detected across 89 epochs, and that two ictal candidate segments were present at 119.0–132.5 s and 136.5–199.0 s. It also restated that no direct channel overlap existed between HFO consensus channels and seizure supportive channels, and that this mismatch lowered confidence. The final impression concluded that no concordance was present between HFO consensus and seizure evidence, and that the hypothesis should therefore remain deferred pending further integration. This indicates that version 4 preserved the cautious multimodal reasoning of version 3 while adding adaptive orchestration capability on top of it.

Version 4 final report for subject HUP146

Case overview

Subject sub-HUP146 from the HUP dataset was analyzed with integrated HFO and seizure branches. Cross-modal comparison used exact channel-name matching.

HFO findings

HFO consensus strength was strong. Strong consensus channels were LG8, RC5. Additional supportive channels were RI11, RJ5, LG7, LG5, LG4. High HFO activity with 246 total FR events and 18 FRANDR pairs detected across 89 epochs.

Seizure findings

Two ictal segments detected at 119-132.5s (22 positive windows, 0.88 peak prob) and 136.5-199s (120 positive windows, 0.91 peak prob). Low interictal false-positive burden. Seizure channel-level

evidence was treated as model-derived saliency proxy. Top supportive seizure channels were RG1-RG2, RG2-RG3, RD3-RD4, RA2-RA3, RA1-RA2. Seizure evidence supported by model-derived saliency proxy with top channels showing high ictal probability.

Cross-modal integration

Concordance mode was `exact_channel_match`. Concordance level was none. Overlap channels: none. No overlap found between HFO channels and seizure supportive channels under exact matching, leading to cross-modal mismatch warning.

Limitations and confidence

Overall confidence was moderate. Strong HFO consensus on LG8 and RC5, and ictal detection with high peak probability in two segments (119-132.5s and 136.5-199s) support confidence. However, cross-modal mismatch warning limits integration confidence. Cross-modal mismatch warning present No exact channel overlap for integration Seizure channel evidence is model-derived saliency proxy rather than direct electrophysiological localization. No direct cross-modal overlap was identified between HFO and seizure supportive channels.

Final impression

Insufficient direct channel overlap and mismatch warning prevent definitive EZ hypothesis.

Figure 4-18: Excerpt from the Version 4 generated report for subject HUP146.

Overall, the main result of version 4 was not a different biological conclusion for sub-HUP146, but the successful demonstration that the system could now inspect the current multimodal case state, generate a validated tool plan, execute bounded tools through Python, and then refresh the downstream reasoning and reporting artifacts. Therefore, version 4 transformed the pipeline from a static reasoning system into an adaptive, orchestrated system. The Table 4.18 shows a summary of the functions and limitations of each version and the improvements made throughout the whole process.

4.3.5 Rubric Evaluation

The final LLM system (version 4) was evaluated on the case-study 5 subjects (sub-HUP126, sub-HUP130, sub-HUP146, sub-HUP157, sub-HUP164) using three complementary criteria: reasoning-quality rubric, faithfulness to structured evidence, and orchestration validity.

Rubric-based reasoning quality The reasoning-quality rubric assessed whether:

Table 4.18: Comparison of LLM system versions for subject sub-HUP146.

Version	Main function	Improvement	Remaining limitation
Version 1	Generated report from compact JSON	Produced first structured report	Weak cross-modal reasoning
Version 2	Added seizure-channel evidence and fusion logic	Improved comparison between HFO and seizure branches	LLM output still variable
Version 3	Added schema validation and deterministic report rendering	Improved reliability and reproducibility	No tool planning
Version 4	Added bounded tool orchestration	Allowed planner-based execution of selected tools	Evaluated on limited case-study data

- Generated reasoning and final report adequately covered the seizure, HFO, and integration findings
- System performed explicit cross-modal comparison
- Uncertainty was handled appropriately
- Constraints were respected
- Output was clear and well structured

Across the five evaluated subjects, the final system achieved a mean total rubric score of 9.6/10. Four subjects (sub-HUP126, sub-HUP130, sub-HUP146, and sub-HUP164) obtained the maximum score of 10/10, while sub-HUP157 obtained a slightly lower score of 8/10. The Table 4.20 shows the detailed rubric scores for all five subjects. The average per-criterion scores were 2.0/2.0 for evidence coverage, 2.0/2.0 for cross-modal reasoning, 1.8/2.0 for uncertainty handling, 2.0/2.0 for constraint adherence, and 1.8/2.0 for clarity and structure.

In terms of evidence coverage, the generated outputs consistently included all major findings from the compact case JSONs, including HFO strong-consensus channels, HFO supportive channels, seizure-supportive channels, ictal segment timing when present, concordance level, overlap status, and the final integration summary. Cross-modal reasoning was also strong across all five cases, because the system did not only mention seizure and HFO findings independently. Instead, it explicitly compared

Table 4.19: Average reasoning-quality results across the five evaluated subjects.

Metric	Average score
Evidence coverage	2.0 / 2
Cross-modal reasoning	2.0 / 2
Uncertainty handling	1.8 / 2
Constraint adherence	2.0 / 2
Clarity and structure	1.8 / 2
Overall mean total score	9.6 / 10

Table 4.20: Reasoning-quality rubric scores for the five evaluated subjects. Scores were assigned as 0 = poor/absent, 1 = partial/acceptable, and 2 = good/complete. Maximum total per subject = 10.

Subject	Evidence	Cross-modal	Uncertainty	Constraint	Clarity	Total
sub-HUP126	2	2	2	2	2	10
sub-HUP130	2	2	2	2	2	10
sub-HUP146	2	2	2	2	2	10
sub-HUP157	2	2	1	2	1	8
sub-HUP164	2	2	2	2	2	10
Average	2.0	2.0	1.8	2.0	1.8	9.6

them and described whether overlap was present or absent, and how this affected the final interpretation.

The uncertainty-handling criterion received high scores overall, although it showed the main variation across subjects. In sub-HUP126, sub-HUP130, sub-HUP146, and sub-HUP164, the system appropriately adjusted confidence according to the integration findings, especially when mismatch warnings and absent concordance were present. In these cases, the reports remained conservative and avoided overly strong localization statements. However, sub-HUP157 received a slightly lower score for uncertainty handling because the report was less conservative in its confidence phrasing despite the presence of mismatch-related limitations.

Constraint adherence was complete across all five subjects. No invented channels, unsupported outcomes, or non-evidence-based conclusions were observed. In addition, seizure-related channel evidence was consistently described as model-derived saliency proxy evidence rather than direct clinical seizure-onset localization. Finally, clarity

and structure were strong in most cases, with clear sections covering seizure findings, HFO findings, integration, limitations, and confidence. The only slight reduction was observed for sub-HUP157, where some report phrasing was less polished and slightly more repetitive than in the other subjects.

Faithfulness to structured evidence In terms of checking the faithfulness of the system, the checklist presented in Table 4.21 and Table 4.22 reveals whether the generated reasoning and final report remained consistent with the structured compact case JSONs. Across the five evaluated subjects, the average faithfulness score was 9.8/10, corresponding to 98%. Four subjects achieved 10/10 (100%), while sub-HUP157 achieved 9/10 (90%).

Overall, the reports preserved the HFO strong-consensus channels, HFO supportive channels, seizure-supportive channels, ictal segment timings, concordance level, and overlap or no-overlap status correctly. The reports also correctly described seizure channel-level evidence as model-derived saliency proxy evidence. No invented channels or unsupported diagnosis, onset location, or outcome statements were observed in any of the five subjects.

The only reduction in faithfulness was observed for sub-HUP157, where the confidence direction was judged to be slightly less aligned with the mismatch and consistency flags than in the other cases. Nevertheless, even in this subject, the report remained largely evidence-bound and preserved the major structured findings correctly.

Table 4.21: Faithfulness summary scores for the five evaluated subjects. The score is the number of checklist items satisfied out of 10.

Subject	Faithfulness score	Faithfulness (%)
sub-HUP126	10/10	100%
sub-HUP130	10/10	100%
sub-HUP146	10/10	100%
sub-HUP157	9/10	90%
sub-HUP164	10/10	100%
Average	9.8/10	98%

Evaluation of tool orchestration Because the evaluated system corresponds to version 4, an additional tool orchestration checklist was applied to assess the bounded

Table 4.22: Faithfulness checklist results across the five evaluated subjects.

Item	HUP126	HUP130	HUP146	HUP157	HUP164
HFO strong consensus channels are reported correctly	Yes	Yes	Yes	Yes	Yes
HFO supportive channels are reported correctly	Yes	Yes	Yes	Yes	Yes
Seizure supportive channels are reported correctly	Yes	Yes	Yes	Yes	Yes
Ictal segment timing is preserved when present	Yes	Yes	Yes	Yes	Yes
Concordance level matches compact case	Yes	Yes	Yes	Yes	Yes
Overlap / no-overlap status matches compact case	Yes	Yes	Yes	Yes	Yes
Seizure evidence is correctly described as model-derived saliency proxy	Yes	Yes	Yes	Yes	Yes
Confidence direction is consistent with mismatch / consistency flags	Yes	Yes	Yes	No	Yes
No invented channels are present	Yes	Yes	Yes	Yes	Yes
No unsupported diagnosis, onset location, or outcome is introduced	Yes	Yes	Yes	Yes	Yes

planner and execution pipeline. All orchestration criteria were satisfied across the five evaluated subjects. As shown in Table 4.24 and Table 4.23, each subject achieved the maximum orchestration validity score of 7/7.

For all five subjects, the planner output was valid JSON, the planning decision was appropriate for the case state, only allowed tool names were used, the action order was valid, tool execution completed successfully, the updated reasoning JSON remained valid, and the final report was regenerated successfully. The appropriateness of the planning decision was assessed using a predefined rule. If the case contained cross-modal mismatch and concordance level none, a rebuild and rerun strategy was considered appropriate. If the case was otherwise incomplete, such as when reasoning or report outputs were missing, rerunning reasoning and report generation was also considered appropriate. Under this rule, all five planning decisions were judged to be valid.

Overall, these results indicate that the final version of the system was able to generate reports that were not only well organized but also faithful to the structured evidence and consistent with the orchestration logic introduced in version 4. The short summary for each criterion is shown in Table 4.25. Across the five evaluated subjects, the system achieved a mean reasoning-quality score of 9.6/10, a mean faithfulness score of 9.8/10 (98%), and a mean orchestration validity score of 7.0/7. These findings suggest that the final system can reliably integrate seizure and HFO evidence

Table 4.23: Orchestration validity summary for the five evaluated subjects. Maximum score per subject = 7.

Subject	Orchestration validity score
sub-HUP126	7/7
sub-HUP130	7/7
sub-HUP146	7/7
sub-HUP157	7/7
sub-HUP164	7/7
Average	7.0/7

Table 4.24: Tool orchestration checklist results for version 4 across the five evaluated subjects.

Item	HUP126	HUP130	HUP146	HUP157	HUP164
Planner output is valid JSON	Yes	Yes	Yes	Yes	Yes
Planning decision is appropriate for the case state	Yes	Yes	Yes	Yes	Yes
Only allowed tool names are used	Yes	Yes	Yes	Yes	Yes
Action order is valid	Yes	Yes	Yes	Yes	Yes
Tool execution completes successfully	Yes	Yes	Yes	Yes	Yes
Updated reasoning JSON is valid	Yes	Yes	Yes	Yes	Yes
Final report is regenerated successfully	Yes	Yes	Yes	Yes	Yes

while remaining constrained to the available evidence and generally conservative in the presence of cross-modal disagreement.

Table 4.25: Overall LLM evaluation summary across the five evaluated subjects.

Subject	Reasoning quality	Faithfulness	Orchestration validity
sub-HUP126	10/10	10/10 (100%)	7/7
sub-HUP130	10/10	10/10 (100%)	7/7
sub-HUP146	10/10	10/10 (100%)	7/7
sub-HUP157	8/10	9/10 (90%)	7/7
sub-HUP164	10/10	10/10 (100%)	7/7
Average	9.6/10	9.8/10 (98%)	7.0/7

Chapter 5

Conclusion

This thesis presented a framework for the automatic generation of text-based summaries of intracranial EEG seizure events by combining signal processing, deep learning, and large language models. The work addressed an important gap identified in the literature: although HFO detection, seizure detection, and LLM-based reporting have each been studied separately, there is still no unified iEEG system that combines these components into one structured reporting pipeline.

The first contribution of the thesis was the development of two complementary analytical branches. For HFO detection, the workflow was first implemented on the OpenNeuro ds003498 dataset and then adapted to the HUP dataset with changes in preprocessing, filtering, and stricter detector settings. The final HFO branch focused on ripple, fast ripple, and FRandR analysis, with percentile-based identification of candidate HFO-area channels. For seizure detection, the work progressed from early experiments on the SWEC-ETHZ dataset to a CNN-transformer model trained on the HUP dataset. Across 43 HUP subjects, the final seizure model achieved mean subject-level accuracy and balanced accuracy of 0.83, with median values of 0.87. The median F1-score was 0.84, while median AUROC and AUPRC reached 0.97 and 0.95, respectively. These results showed that the seizure branch was effective overall, although performance remained heterogeneous across subjects.

The second contribution was the design of the LLM-based reporting framework. The system evolved through four versions, starting from a simple report generator and

progressing toward a more constrained reasoning system with deterministic integration, schema validation, and bounded tool orchestration. In the final version, the LLM did not analyze raw iEEG directly. Instead, it reasoned over structured JSON outputs from the HFO and seizure branches, generated an intermediate reasoning representation, and produced a final report in a structured clinical style. This design improved control over the output and reduced the risk of unsupported free-form generation.

The final LLM evaluation showed that the framework performed strongly on the five evaluated HUP subjects. The system achieved high scores in reasoning quality, faithfulness to structured evidence, and orchestration validity. In most cases, it correctly summarized HFO findings, seizure findings, and integration-level interpretation, preserved the channel-level and timing information from the structured inputs, and avoided unsupported claims. The strongest aspects of the final system were evidence coverage, cross-modal reasoning, and constraint adherence. These findings suggest that the final framework can generate evidence-grounded iEEG reports while remaining largely faithful to the structured outputs of the analytical modules.

At the same time, several limitations remain. First, both HFO and seizure analysis depend strongly on dataset properties, and the seizure branch still showed subject-to-subject variability. Second, seizure channel evidence used by the LLM was model-derived saliency proxy evidence rather than direct electrophysiological onset labeling, so it must be interpreted cautiously. Third, confidence phrasing in the final narrative was not perfectly consistent across all ambiguous cases. Finally, concordance between HFO and seizure findings was sensitive to channel representation, especially when seizure evidence was expressed as bipolar channel pairs and HFO evidence as single contacts. This shows that deterministic integration logic remains a critical part of the system.

Overall, this thesis demonstrates that automatic iEEG report generation can be approached as a structured multimodal pipeline rather than as free-form text generation. By combining HFO detection, seizure detection, deterministic fusion, and constrained LLM reasoning, the proposed framework provides a step toward more automated and

standardized iEEG reporting. Future work should focus on improving seizure-model robustness, refining concordance logic, expanding validation on larger cohorts, and evaluating the generated reports with clinicians in real reporting workflows.

A.1 Subject sub-HUP126

Version 4 final report for subject HUP126

Case overview

Subject sub-HUP126 from the HUP dataset was analyzed with integrated HFO and seizure branches. Cross-modal comparison used exact channel-name matching.

HFO findings

HFO consensus strength was moderate. Strong consensus channels were LDAH4. Additional supportive channels were LDMH2, LDMH1, LDMH3, LDMH4, LDAH2, LDA1, LDAH3, LDAH1, LDA2, LDAH5. 58 epochs, 543 ripples, 179 FR events, 129 FR+DR pairs detected with moderate consensus strength.

Seizure findings

Ictal activity detected 120.0-159.0 sec with 73 positive windows (mean prob 0.886, peak 0.937) Low interictal false-positive burden Seizure channel-level evidence was treated as model-derived saliency proxy. Top supportive seizure channels were LDAH4-LDAH5, LDMH3-LDMH4, LDAH1-LDAH2, LDAH2-LDAH3, LDMH2-LDMH3. Seizure evidence is proxy-derived with strong cross-modal overlap.

Cross-modal integration

Concordance mode was contact_overlap_from_bipolar_pairs. Concordance level was strong. Overlap channels: LDAH4, LDAH5, LDMH3, LDMH4, LDAH1, LDAH2, LDAH3, LDMH2. Strong cross-modal contact overlap from bipolar pair expansion.

Limitations and confidence

Overall confidence was moderate. HFO consensus strength (moderate) and strong overlap/concordance (strong) provide key support, though seizure evidence remains proxy-derived. Seizure channel evidence is model-derived saliency proxy rather than direct electrophysiological localization. Cross-modal interpretation should be treated cautiously. Findings should be interpreted in the context of model and integration limitations.

Final impression

Supported by strong HFO consensus and cross-modal overlap, but seizure evidence remains proxy-derived.

Figure A-1: Excerpt from the Version 4 generated report for subject HUP126.

A.2 Subject sub-HUP130

Version 4 final report for subject HUP130

Case overview

Subject sub-HUP130 from the HUP dataset was analyzed with integrated HFO and seizure branches. Cross-modal comparison used exact channel-name matching.

HFO findings

HFO consensus strength was moderate. Strong consensus channels were LA1. Additional supportive channels were RB2, RA1, RB3. HFO consensus shows moderate strength with LA1 as strong consensus and RB2, RA1, RB3 as supportive channels.

Seizure findings

Two ictal candidate segments detected: 126.0-170.0s (44s) and 176.0-181.0s (5s) with peak probabilities 0.916 and 0.452 respectively. Interictal false-positive burden is low. Seizure channel-level evidence was treated as model-derived saliency proxy. Top supportive seizure channels were RA1-RA2, RD1-RD2, LG2-LG3, RF3-RF4, LB4-LB5. Seizure evidence is proxy-derived with top supportive channels RA1-RA2, RD1-RD2, LG2-LG3, RF3-RF4, LB4-LB5.

Cross-modal integration

Concordance mode was exact_channel_match. Concordance level was none. Overlap channels: none. Concordance level is none due to no direct overlap between HFO evidence and seizure supportive channels under exact matching. Cross-modal mismatch warning present.

Limitations and confidence

Overall confidence was limited. HFO consensus strength is moderate with strong consensus on LA1 and supportive channels RB2, RA1, RB3. Integration flags indicate cross-modal mismatch due to no direct overlap between HFO evidence and seizure supportive channels under exact matching. Cross-modal mismatch warning present. Seizure channel evidence is model-derived saliency proxy rather than direct electrophysiological localization. No direct cross-modal overlap was identified between HFO and seizure supportive channels. Strong or moderate HFO evidence is present, but no direct overlap with seizure supportive channels was found under exact matching.

Final impression

Insufficient overlap and mismatch warning prevent definitive EZ hypothesis.

Figure A-2: Excerpt from the Version 4 generated report for subject HUP130.

A.3 Subject sub-HUP157

Version 4 final report for subject HUP157

Case overview

Subject sub-HUP157 from the HUP dataset was analyzed with integrated HFO and seizure branches. Cross-modal comparison used exact channel-name matching.

HFO findings

HFO consensus strength was moderate. Strong consensus channels were none. Additional supportive channels were LA10, LA9, LA1, RB2, RB1, LB1, LH8. 38 epochs, 123 ripples, 41 fast ripples, 9 frandr pairs.

Seizure findings

Ictal segments detected at [15-18s], [30-38s], [39-58s] with peak probs 0.256-0.389. Low interictal false-positive burden. Seizure channel-level evidence was treated as model-derived saliency proxy. Top supportive seizure channels were RA1-RA2, LC8-LC9, RA2-RA3, RC2-RC3, LA2-LA3. Supportive channels include RA1-RA2, LC8-LC9, RA2-RA3, RC2-RC3, LA2-LA3.

Cross-modal integration

Concordance mode was exact_channel_match. Concordance level was none. Overlap channels: none. No overlap found under exact matching, cross-modal mismatch warning present.

Limitations and confidence

Overall confidence was moderate. HFO consensus strength (moderate) and seizure evidence proxy type (model-derived saliency proxy) with limited overlap/concordance. Cross-modal mismatch warning Limited overlap/concordance Seizure channel evidence is model-derived saliency proxy rather than direct electrophysiological localization. No direct cross-modal overlap was identified between HFO and seizure supportive channels.

Final impression

No direct overlap with seizure channels under exact matching.

Figure A-3: Excerpt from the Version 4 generated report for subject HUP157.

A.4 Subject sub-HUP164

Version 4 final report for subject HUP164

Case overview

Subject sub-HUP164 from the HUP dataset was analyzed with integrated HFO and seizure branches. Cross-modal comparison used exact channel-name matching.

HFO findings

HFO consensus strength was strong. Strong consensus channels were LG8, RC5. Additional supportive channels were RI11, RJ5, LG7, LG5, LG4. Strong HFO consensus with high ripple and FR rates across key channels.

Seizure findings

Ictal activity detected from 120.5-195.0 sec with peak probability 0.571, supported by model-derived saliency proxy. Moderate interictal false-positive burden noted. Seizure channel-level evidence was treated as model-derived saliency proxy. Top supportive seizure channels were RB11-RB12, RC2-RC3, RC1-RC2, LK4-LK5, LK3-LK4. Seizure evidence supported by model-derived saliency proxy across multiple channels.

Cross-modal integration

Concordance mode was exact_channel_match. Concordance level was none. Overlap channels: none. No direct overlap between HFO and seizure channels under exact matching, with cross-modal mismatch and moderate false-positive burden.

Limitations and confidence

Overall confidence was limited. HFO consensus strength is strong across multiple channels, but integration flags indicate cross-modal mismatch and moderate interictal false-positive burden. Cross-modal mismatch warning Moderate interictal false-positive burden Seizure channel evidence is model-derived saliency proxy rather than direct electrophysiological localization. No direct cross-modal overlap was identified between HFO and seizure supportive channels.

Final impression

Insufficient direct overlap evidence to support epileptogenic zone hypothesis.

Figure A-4: Excerpt from the Version 4 generated report for subject HUP164.

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